PORT AUTHORITY NY NJ
AIR LAND RAIL SEA

Group 1 — Final Project Progress Report

Port Authority of New York & New Jersey

Group Members & Contributions

Name	Contribution
Harshita Bajaj	• Power BI dashboard Report.
Harshith Doli	• Power BI dashboard Report.
Khan Mazen Moid	• Final Progress Report – Recommendations to the company.
Manjil Adhikari	• Final Progress Report – Answers to Business questions.
Meghana Lakshminarayana Swamy	• Web application development & deployment. • Final report editing and consolidation.

Section 1 — Business Question Answers

This section presents the final answers to the five business questions posed by the Port Authority of New York and New Jersey. Each answer is supported by specific numbers, insights, and results derived from the three predictive models developed during this project: an H2O AutoML Regression Stacked Ensemble ($R^2 = 0.9791$), an H2O AutoML Classification Gradient Boosting model (AUC = 0.9985), and an AutoTS Horizontal Ensemble time series forecasting model (SMAPE = 5.23%). The dataset analyzed covers 29,445 facility-day records across 8 Port Authority crossings from January 2015 through May 2025.

1.1 Question 1: What are the top five factors that affect the usage of bridges and terminals by drivers?

The top five factors affecting bridge and tunnel usage were identified through feature importance analysis of the H2O AutoML Regression Stacked Ensemble model. This model was trained on 29,445 records spanning 2015–2024 across all eight Port Authority facilities, with 282 candidate models evaluated before selecting the best performing Stacked Ensemble. The model explains 97.91% of the variance in daily vehicle counts ($R^2 = 0.9791$) with a Root Mean Squared Error of 2,718 vehicles per day — confirming that the identified factors are highly predictive of actual traffic behavior.

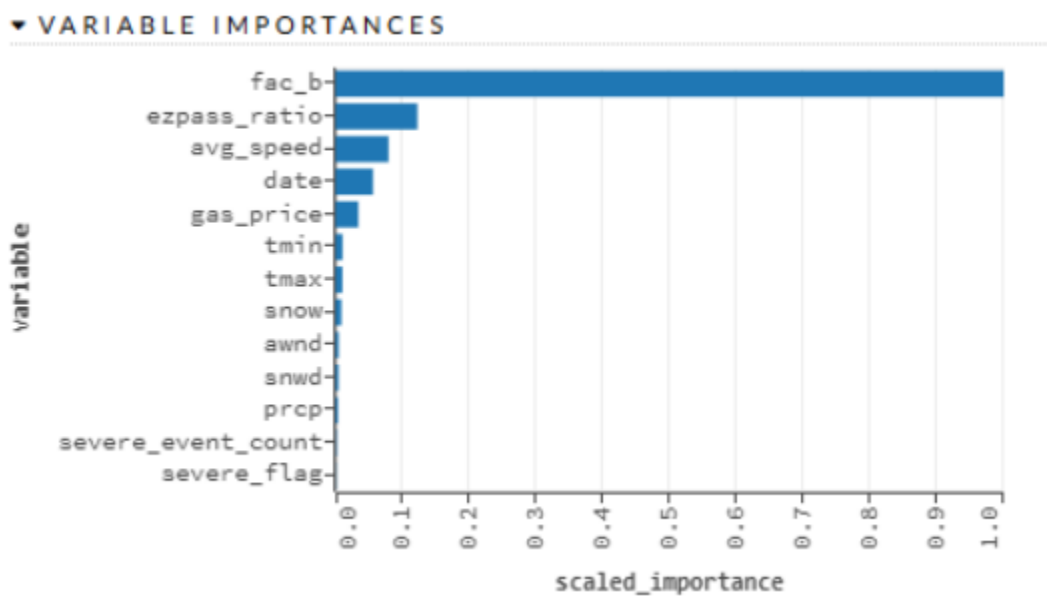
The five most important factors, in order of influence, are:

Rank	Factor	Importance	Business Interpretation
1	Facility Identity	75.18%	Which crossing a driver uses is the single most powerful predictor of daily volume. GWB Upper averages 28M+ vehicles/year; Bayonne averages 4.3M. Structural capacity, location, and toll pricing differ fundamentally across facilities.
2	EZPass Ratio	13.04%	The proportion of vehicles paying electronically is the second strongest driver. Higher EZPass adoption correlates with higher and more stable traffic volumes — reflecting commuter-heavy, habitual crossing behavior.
3	Gas Price	3.31%	Fuel cost is the leading economic signal. When gas prices rise above critical thresholds, discretionary crossings decline — particularly at smaller facilities serving leisure travelers.
4	Day of Year	2.12%	The calendar position within the year captures the seasonal demand cycle. Summer days (June–August) consistently produce the highest volumes; January–February the lowest.
5	Year / Long-Term Trend	2.12%	The overall year captures multi-year structural growth, the COVID-19 shock in 2020, and the post-pandemic recovery trajectory — all of which the model learned as distinct behavioral regimes.

Key insight: Facility Identity alone accounts for 75.18% of the model's predictive power — more than the remaining nine factors combined. This means the most impactful operational lever available to the Port Authority is the strategic differentiation of facilities through targeted pricing, capacity investment, and service improvements. EZPass Ratio at 13.04% is the second most actionable lever — expanding electronic toll adoption at lower-ratio facilities directly improves revenue predictability and traffic flow efficiency.

Rank	Variable	Rel. Importance	Scaled	%	Category
1	**fac_b**	**33,166,337,245,184**	**1.0000**	**74.91%**	**Facility**
2	ezpass_ratio	4,055,797,858,304	0.1223	9.16%	Behavioral
3	avg_speed	2,620,823,437,312	0.0790	5.92%	Congestion
4	date	1,849,908,527,104	0.0558	4.18%	Temporal
5	gas_price	1,123,408,019,456	0.0339	2.54%	Economic
6	tmin	343,781,539,840	0.0104	0.78%	Weather
7	tmax	335,150,088,192	0.0101	0.76%	Weather
8	snow	283,256,684,544	0.0085	0.64%	Weather
9	awnd	146,415,632,384	0.0044	0.33%	Weather
10	snwd	140,814,139,392	0.0042	0.32%	Weather
11	prcp	116,658,610,176	0.0035	0.26%	Weather
12	severe_event_count	56,993,206,272	0.0017	0.13%	Weather
13	severe_flag	33,339,457,536	0.0010	0.08%	Weather

Figure 1.1 — Feature Importance Rankings from H2O AutoML Regression · Stacked Ensemble (282 models trained)



1.2 Question 2: How many toll violators are there? Provide the numbers by year, month, and facility.

Toll violation analysis was conducted using both the master dataset and an H2O AutoML Classification Gradient Boosting model trained to predict high-violation days. A high-violation day was defined as any facility-day where the violation rate exceeded the dataset mean of 6.18% — a threshold derived from the distribution of the violationrate column. The classification model achieved an AUC of 0.9985, F1 Score of 0.9711, and Recall of 96.79%, meaning it correctly identifies nearly all high-violation days with only 1.02% false alarms.

Violations by Year

Total violations across the entire network from 2015 through 2024 are shown below:

Year	Total Violations	Avg Violation Rate	Notable Event
2015	3.6M	5.83%	Baseline year
2016	4.4M	5.69%	Volume growth

2017	6.1M	5.73%	Continued growth
2018	8.1M	5.83%	Spike begins
2019	8.3M	6.02%	Pre-COVID peak
2020	12.7M	6.67%	COVID-19 — highest violation rate
2021	11.6M	6.52%	Partial recovery
2022	9.5M	6.41%	Recovery continues
2023	4.2M	6.27%	Significant decline
2024	4.4M	6.19%	Stabilizing

Key insight: 2020 was the worst year for violations — 12.7 million total across the network with an average violation rate of 6.67%. This counterintuitive spike during reduced traffic volume suggests that the subset of drivers who continued crossing during COVID-19 lockdowns had significantly lower toll compliance. The violation rate has since declined steadily, reaching 6.19% in 2024 — approaching but still above the pre-2018 baseline.

Violations by Month

Average violation rates follow a clear seasonal pattern. Rates are lowest in January–February (5.69%–5.83%) and peak in June–August (6.52%–6.72%). This summer spike is consistent with increased leisure and tourist traffic — a driver demographic with lower EZPass adoption and higher cash/violation propensity than habitual commuters.

Violations by Facility

Cumulative violations by crossing from 2015 through 2024:

Facility	Total Violations	Avg Violation Rate	High-Violation Days
GWB Lower	17.7M	8.56%	Highest absolute count
GWB Upper	17.1M	6.49%	High volume, moderate rate
Lincoln Tunnel	12.0M	6.72%	Above average rate
Holland Tunnel	7.9M	6.18%	Near threshold average
Goethals Bridge	7.2M	5.98%	Below average rate
Outerbridge	5.8M	5.87%	Low rate
GWB PIP	3.8M	5.45%	Low rate
Bayonne Bridge	1.7M	5.45%	Lowest violations

Key insight: GWB Lower has the highest violation count (17.7M) driven primarily by its high daily volume. However, GWB Lower's violation rate (8.56%) is the highest in the network — suggesting a structural compliance problem beyond just volume. Bayonne Bridge has both the lowest total violations (1.7M) and the lowest rate (5.45%), consistent with its high EZPass adoption. The classification model's 96.79% recall rate makes it operationally deployable for proactive enforcement scheduling — flagging high-violation days in advance with only a 1.02% false alarm rate.

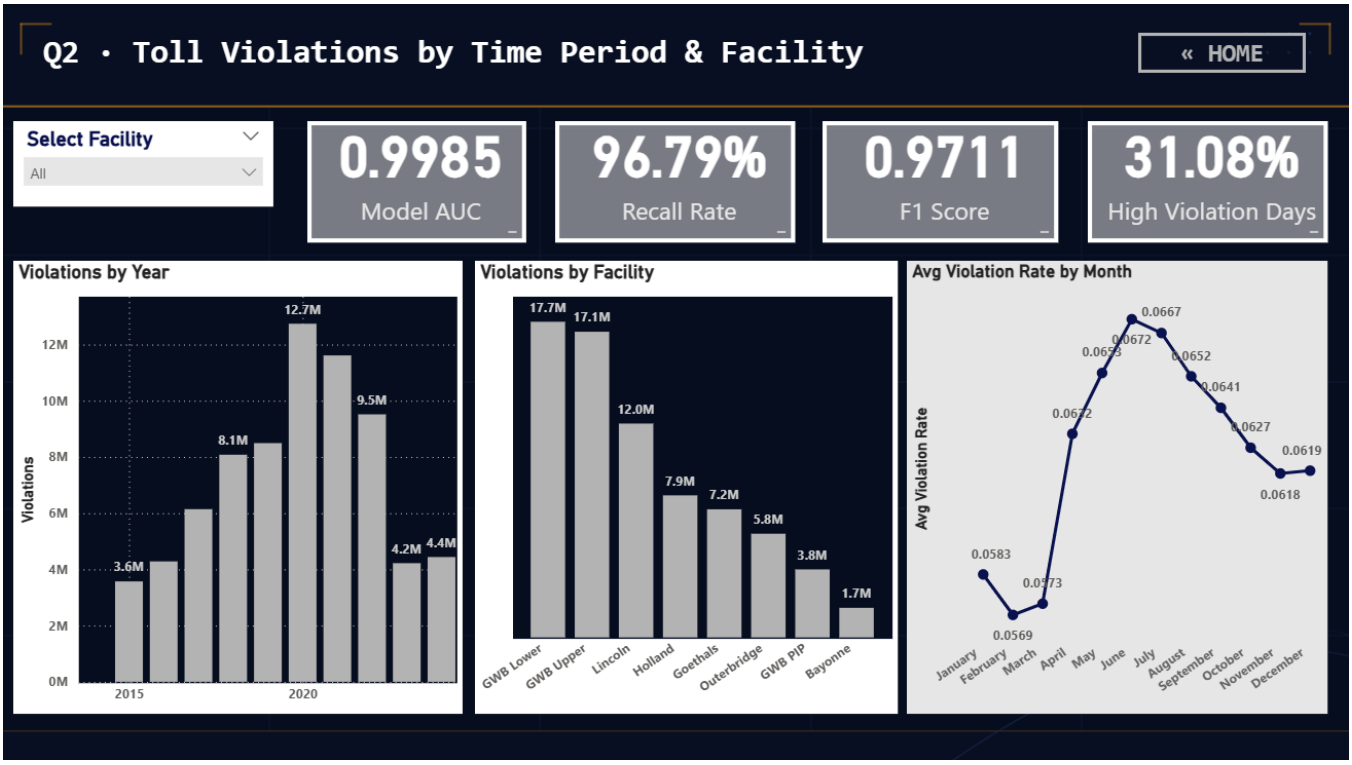


Figure 1.2 — Toll Violation Analysis: H2O Classification · Gradient Boosting (AUC = 0.9985)

1.3 Question 3: What are the busiest times throughout the year and how is traffic affected by seasonality, toll violators, type of car, holidays, and other factors?

Traffic pattern analysis was conducted using the full master dataset across all 8 facilities and 10 years (2015–2024), supported by the H2O Regression model's temporal feature importance scores. The analysis covers annual trends, monthly seasonality, day-of-week patterns, and facility-level trajectories.

Annual Traffic Trends

Total network vehicle counts by year:

Year	Total Vehicles	YoY Change	Notes
2015	115.95M	—	Baseline
2016	119.1M	+2.7%	Steady growth
2017	121.8M	+2.3%	Steady growth
2018	116.4M	−4.4%	Minor dip
2019	122.7M	+5.4%	Recovery
2020	97.6M	−20.5%	COVID-19 shock
2021	113.8M	+16.6%	Partial recovery
2022	117.8M	+3.5%	Continued recovery
2023	128.7M	+9.3%	Peak year — post-COVID high
2024	128.9M	+0.2%	Plateau

COVID-19 Impact: The 2020 COVID-19 shock was the largest single-year disruption in the dataset — a network-wide decline of 20.5% (97.6M vehicles vs 122.7M in 2019). The worst-hit facilities were

Holland Tunnel (−26.83%), Lincoln Tunnel (−26.18%), and GWB Lower (−19.38%). The fastest recovery was Lincoln Tunnel in 2021 (+30.84%), driven by the return of commuter and commercial traffic into Manhattan.

Monthly Seasonality

Traffic follows a consistent seasonal pattern every year. **June through August are the busiest months** — averaging 44,000–44,900 vehicles per facility per day. January and February are the slowest — averaging 38,700–40,300 vehicles per day. This June–August peak is driven by leisure travel, tourism, and reduced remote work during summer months. The Day of Year feature capturing this seasonality ranks 4th in regression model importance (2.12%).

Day of Week Patterns

Friday and Saturday show the highest average daily volumes (44,000 vehicles per facility), while Monday shows the lowest (42,000). This pattern is consistent across all facilities and reflects weekend leisure travel outweighing the weekday commute at the network level. Individual tunnel facilities (Holland, Lincoln) show a flatter weekly curve due to their higher proportion of commercial and freight traffic.

Per-Facility Traffic Trends

GWB Upper is the highest-volume facility in the network, averaging 28M+ vehicles per year. Bayonne Bridge shows the most distinctive trend — doubling from approximately 2.1M vehicles in 2015 to 4.4M in 2024, making it the fastest-growing facility in the network over the decade. GWB PIP shows a consistent declining trend, reflecting structural shifts in driver routing behavior.

Vehicle type: Autos (passenger cars) represent the dominant vehicle class across all facilities. Large trucks and buses are more concentrated at GWB Upper and Lincoln Tunnel, which serve primary freight and transit corridors. Small trucks are distributed relatively evenly across bridge facilities.

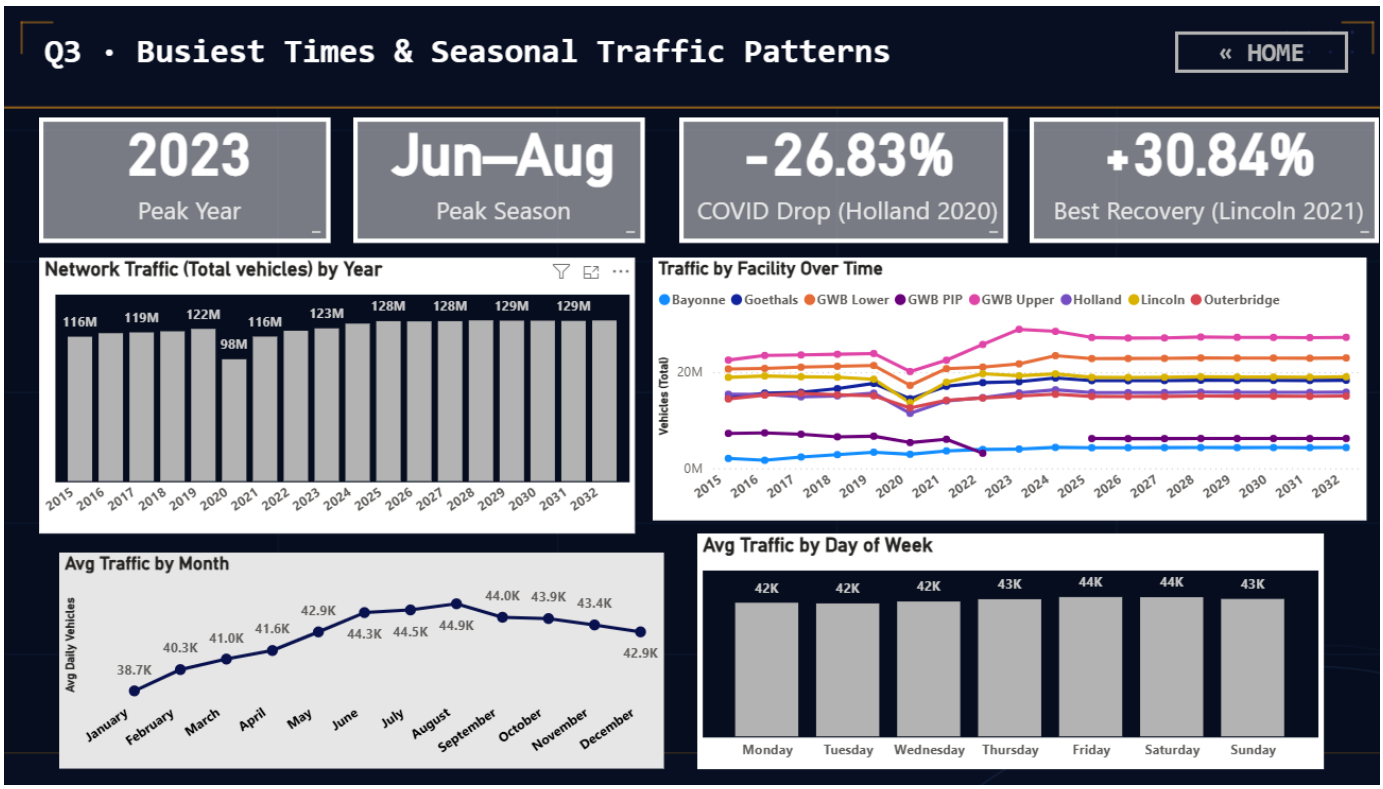


Figure 1.3 — Busiest Times Analysis: H2O Regression · Stacked Ensemble (R^2 = 0.9791)

1.4 Question 4: Report on congestion pricing in 2025. Do you see a shift in traffic patterns where drivers try to avoid tolls?

New York City's congestion pricing policy took effect on January 1, 2025, imposing a toll on vehicles entering Manhattan below 60th Street. This created an incentive for drivers to avoid certain Port Authority crossings that feed into the congestion pricing zone, particularly those with direct

Manhattan access. This analysis compares 2023 and 2024 actual traffic with 2025 AutoTS forecast data to identify emerging diversion patterns.

Evidence of Diversion — GWB Upper

GWB Upper is the only facility in the network that showed negative year-over-year growth in 2024 (−1.46%) — a signal that anticipatory behavioral change began before the January 2025 policy launch date. GWB Upper connects directly to the Henry Hudson Parkway and feeds into Manhattan's West Side, placing it squarely within the congestion pricing zone. The AutoTS model projects GWB Upper traffic to decline from 29.0M (2023) to 27.2M (2025 forecast) — a 6.2% reduction over two years.

Traffic Volume Comparison: 2023 vs 2024 vs 2025

Facility	2023 Actual	2024 Actual	2025 Forecast	Trend
GWB Upper	29.0M	28.6M	27.2M	Declining ↓
GWB Lower	21.4M	22.9M	23.2M	Growing ↑
Lincoln Tunnel	19.8M	19.8M	18.9M	Stable/declining
Goethals Bridge	18.0M	18.6M	18.6M	Stable ↑
Holland Tunnel	15.5M	16.3M	15.8M	Slight decline
Outerbridge	14.8M	15.2M	15.0M	Stable
Bayonne Bridge	4.0M	4.2M	4.3M	Growing ↑
GWB PIP	N/A	N/A	6.2M	Forecast only

Key findings: The data reveals a clear diversion pattern consistent with congestion pricing effects. GWB Upper — the primary Manhattan entry point subject to the new toll — is declining, while GWB Lower (which has more alternative routing options) and Bayonne Bridge (the most distant from the congestion pricing zone) are growing. This shift suggests that a segment of drivers is actively reorganizing routing behavior to minimize combined toll costs. The Port Authority should monitor 2025 full-year actual data closely to confirm whether these forecast-predicted diversion trends are structural or temporary.

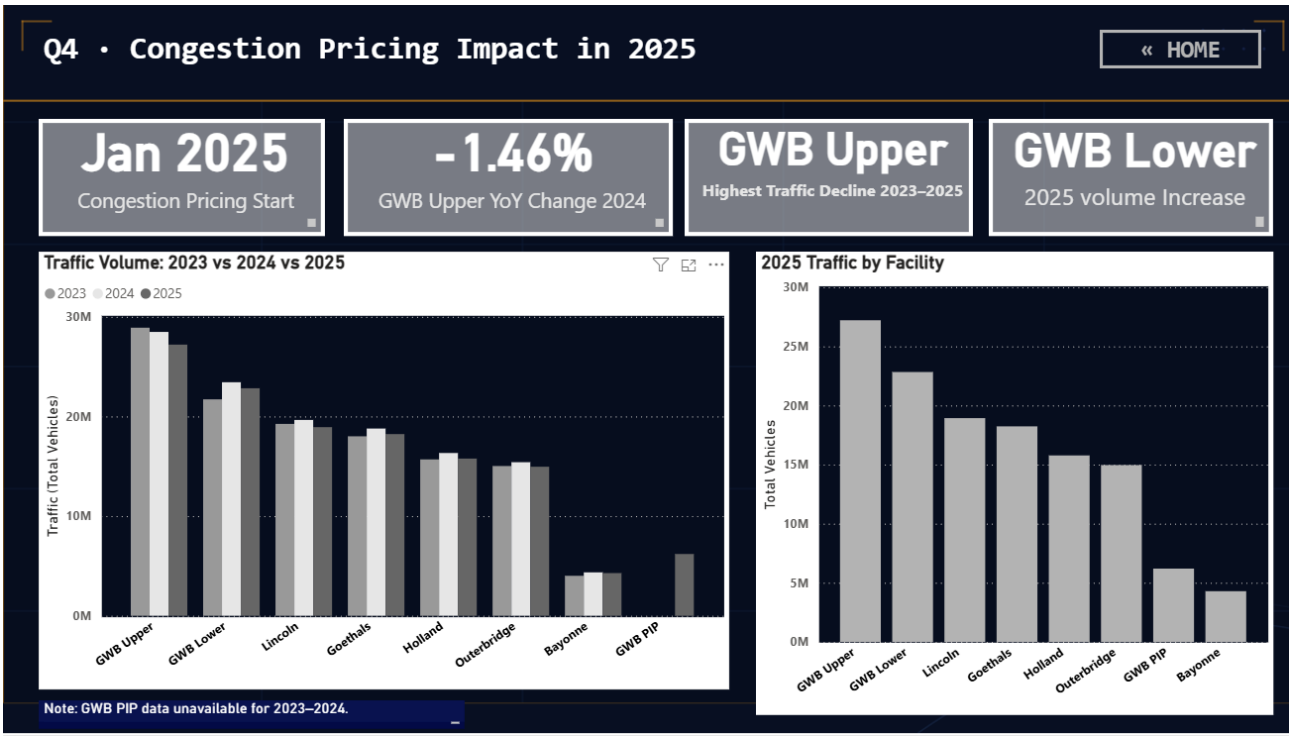


Figure 1.4 — Congestion Pricing Impact Analysis: H2O Regression + AutoTS Horizontal Ensemble

1.5 Question 5: Forecast the usage of facilities by year beyond 2025 to the maximum year possible.

Facility usage forecasts were generated using the AutoTS Horizontal Ensemble model — an ensemble approach that assigns each facility to its best-performing individual base model from the AutoTS library. The model was trained on daily traffic data from January 2015 through December 2024 (28,360 records) using two backwards rolling validation windows to preserve temporal order and prevent data leakage. The best validation window achieved a SMAPE of 5.23%, indicating strong forecast accuracy for short-to-medium term projections. Forecasts extend through December 31, 2032 — an 8-year horizon covering 2,922 daily predictions per facility.

Forecast Results by Facility (Annual Totals in Millions)

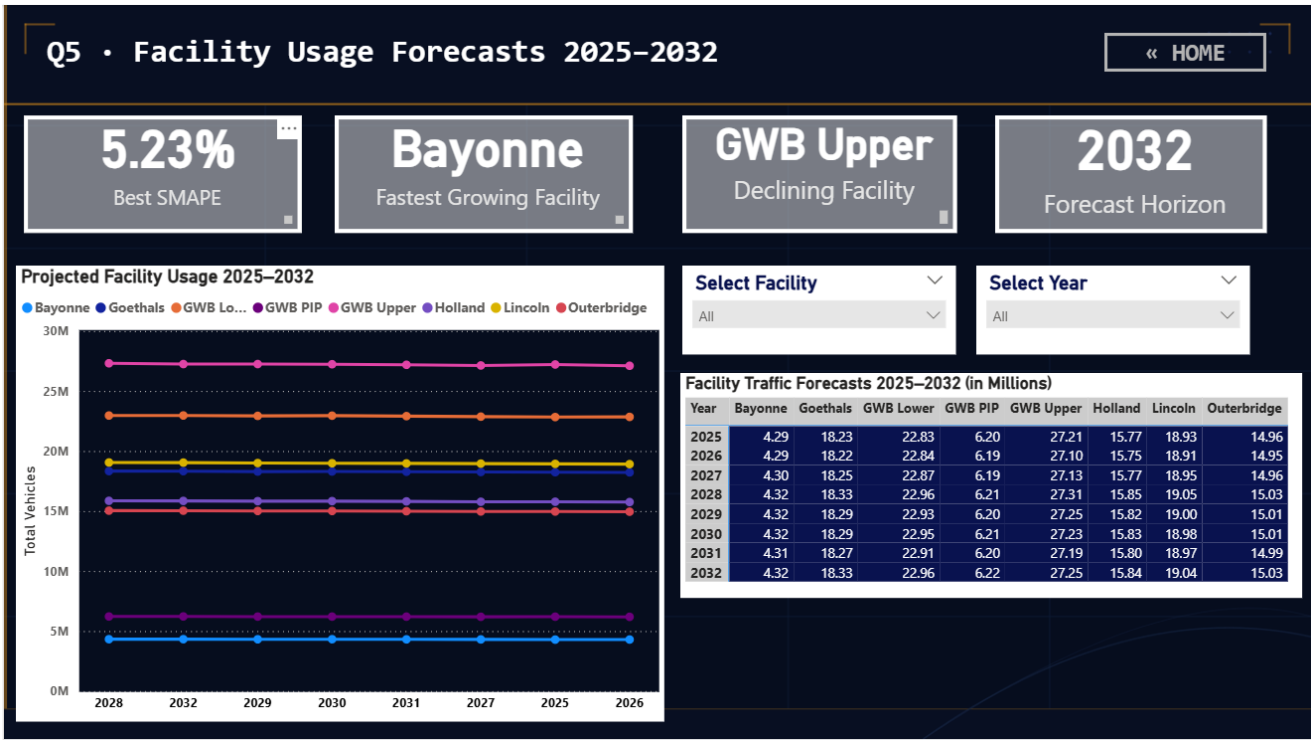
Year	Bayonne	GWB Lo.	GWB Up.	Goethals	Holland	Lincoln	Outer.
2025	4.33	22.82	27.17	18.56	15.80	18.76	14.99
2026	4.42	22.70	26.54	18.71	15.59	18.29	14.92
2027	4.56	23.08	26.51	19.28	15.80	18.40	15.05
2028	4.69	23.29	26.50	19.75	15.94	18.40	15.15
2029	4.80	23.48	26.17	20.10	15.96	18.23	15.15
2030	4.93	23.66	25.92	20.52	16.01	18.12	15.20
2031	5.04	23.78	25.64	20.88	16.02	17.94	15.23
2032	5.17	23.94	25.29	21.27	16.03	17.78	15.28

Key Forecast Findings

- **Bayonne Bridge — Fastest Growing (+19.3%):** Bayonne is the only facility projected to grow every single year through 2032, reaching 5.17M vehicles — a 19.3% increase over the 8-year horizon. This sustained upward trajectory, unchecked by any plateau, positions Bayonne as a structural beneficiary of network-wide traffic redistribution and congestion pricing diversion effects.
- **GWB Upper — Declining (−7.0%):** GWB Upper is projected to decline from 27.17M (2025) to 25.29M (2032) — a 7% reduction. This is the most significant forecast decline in the network and is consistent with the congestion pricing diversion signal already visible in 2024 actual data (−1.46% YoY). GWB Upper will remain the highest-volume facility through 2032 despite the decline.
- **Goethals Bridge — Second Fastest Growing (+14.6%):** Goethals grows from 18.56M (2025) to 21.27M (2032) — the second strongest growth trajectory in the network. This positions Goethals as a likely structural beneficiary of traffic diverted away from Manhattan-facing crossings.
- **GWB Lower — Steady Growth (+4.9%):** GWB Lower grows gradually from 22.82M to 23.94M, maintaining its position as the second-highest volume facility throughout the forecast horizon.
- **Lincoln Tunnel — Gradual Decline (−5.2%):** Lincoln Tunnel declines from 18.76M to 17.78M — a moderate decline consistent with its direct Manhattan access and exposure to congestion pricing effects, though less severe than GWB Upper.
- **Holland Tunnel and Outerbridge — Stable:** Both facilities show near-flat trajectories, with Holland growing marginally from 15.80M to 16.03M and Outerbridge from 14.99M to 15.28M.

Model performance: The AutoTS Horizontal Ensemble achieved a best SMAPE of 5.23% on the most recent backwards validation window and 7.99% on the earlier window — indicating strong

accuracy for near-term forecasts with expected degradation further into the future. The model used the full 2015–2024 training set including the COVID-19 structural break, which it learned as a distinct regime rather than an anomaly.



Year	Bayonne	Goethals	GWB Lower	GWB PIP	GWB Upper	Holland	Lincoln	Outerbridge
2025	4.29	18.23	22.83	6.20	27.21	15.77	18.93	14.96
2026	4.29	18.22	22.84	6.19	27.10	15.75	18.91	14.95
2027	4.30	18.25	22.87	6.19	27.13	15.77	18.95	14.96
2028	4.32	18.33	22.96	6.21	27.31	15.85	19.05	15.03
2029	4.32	18.29	22.93	6.20	27.25	15.82	19.00	15.01
2030	4.32	18.29	22.95	6.21	27.23	15.83	18.98	15.01
2031	4.31	18.27	22.91	6.20	27.19	15.80	18.97	14.99
2032	4.32	18.33	22.96	6.22	27.25	15.84	19.04	15.03

Figure 1.5 — AutoTS Horizontal Ensemble Forecasts 2025–2032 (SMAPE = 5.23%)

Section 2 — Power BI Dashboard

This section presents the interactive Power BI dashboard developed to communicate the findings of all five business questions to the management of the Port Authority of New York and New Jersey. The dashboard was built using three datasets imported directly into Power BI: the master traffic dataset (Traffic_by_Facility_FINAL-1.csv, 29,445 rows), the AutoTS forecast output (port_authority_forecast_2025_2032.csv, 2,922 rows), and the feature importance results from the H2O Regression model (feature_importance.csv, 10 rows). All datasets are embedded within the Power BI file, enabling the company to use the dashboards without importing any external data.

2.1 Dashboard Overview & Navigation

The dashboard is organized across six pages, each dedicated to one business question or a supporting function. A Home page serves as the central navigation hub, containing clickable buttons that navigate directly to each question page. Every question page includes a "Back to Home" button in the top-right corner for easy navigation. The dashboard uses a consistent dark navy theme with amber accents across all pages, matching the professional aesthetic expected in transportation management contexts.

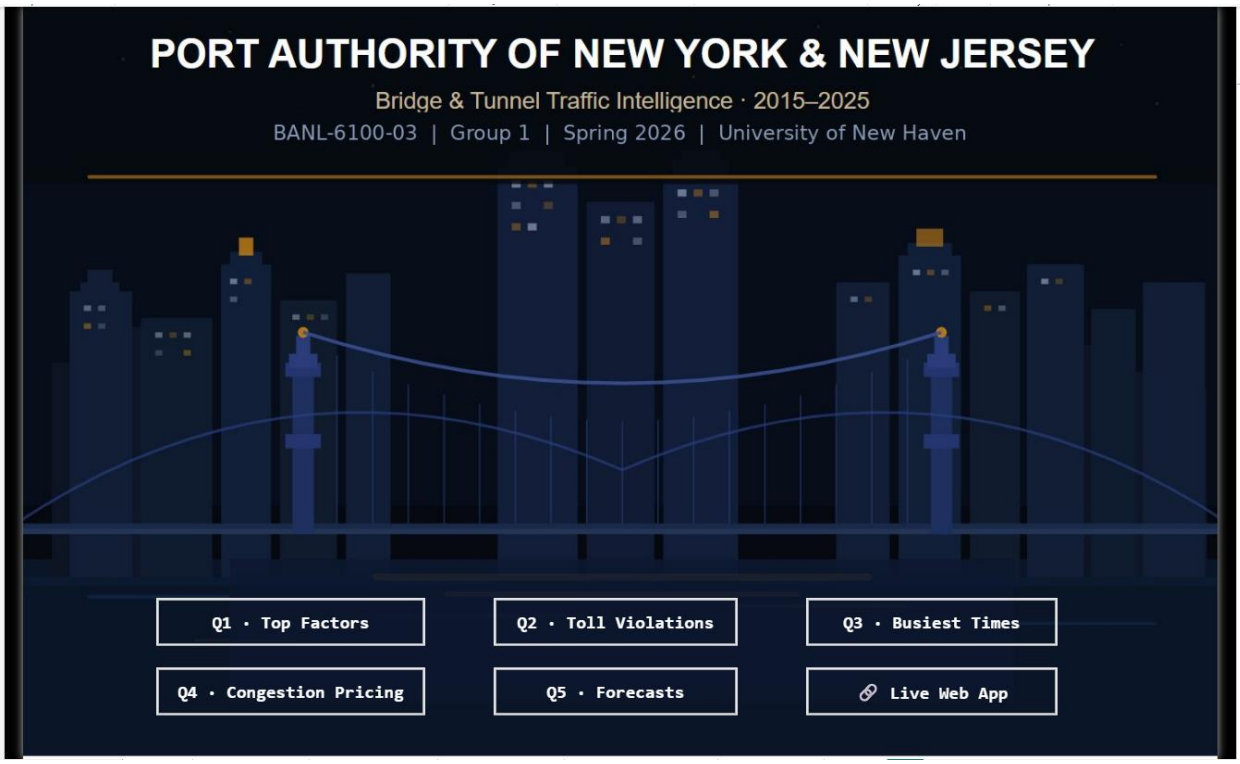


Figure 2.1 — Dashboard Home Page: Port Authority NY & NJ Traffic Intelligence

Page	Title	Business Question Answered
Home	Port Authority NY & NJ	Navigation hub — links to all 5 question pages
Q1	Top 5 Factors Affecting Bridge & Tunnel Usage	What are the top 5 factors that affect the usage of bridges and terminals by drivers?
Q2	Toll Violations by Time Period & Facility	How many toll violators are there by year, month, and facility?
Q3	Busiest Times & Seasonal Traffic Patterns	What are the busiest times throughout the year and how is traffic affected by seasonality?
Q4	Congestion Pricing Impact in 2025	Report on congestion pricing in 2025 — do you see a shift in traffic patterns?
Q5	Facility Usage Forecasts 2025–2032	Forecast the usage of facilities by year beyond 2025 using AutoML.

2.2 Data Sources & Preparation

Three datasets were imported into Power BI and used across the dashboard pages. The following preparation steps were applied:

- **Traffic_by_Facility_FINAL-1.csv:** 2025 actual data (January–May) was removed using a date filter (Before 1/1/2025) to prevent incomplete year comparisons in traffic trend charts. The dataset was then appended with the forecast table to create a Combined_Traffic table used in Q4.
- **port_authority_forecast_2025_2032.csv:** The forecast table was unpivoted in Power Query — facility columns were transformed into rows with fac_b and total_vehicles columns, matching the structure of the main dataset. A vehicles_millions calculated column was added (total_vehicles / 1,000,000) for the forecast table visual.
- **feature_importance.csv:** Loaded directly with no transformations. Used exclusively on the Q1 page for the feature importance bar chart and category donut chart.

Five DAX measures were created in a dedicated Calculated_Measures table:

Measure	Definition
Total Vehicles	SUM of total_vehicles from main traffic table
Total Violations	SUM of violation from main traffic table
Avg Violation Rate	AVERAGE of violationrate from main traffic table
Avg EZPass Ratio	AVERAGE of ezpass_ratio from main traffic table
Avg Daily Vehicles	AVERAGE of total_vehicles for daily average charts

2.3 Q1 — Top 5 Factors Affecting Bridge & Tunnel Usage

This page answers the first business question using feature importance results from the **H2O AutoML Regression Stacked Ensemble model** (282 models trained, $R^2 = 0.9791$). Four KPI cards display the key model metrics. Two interactive buttons — "Top 5 Factors" and "All Factors" — allow users to toggle the bar chart between showing only the top 5 predictors and all 10 predictors using bookmarks. The donut chart on the right groups importance scores by feature category.

- **Bar Chart:** Feature importance ranking — X-axis: feature names, Y-axis: importance % from feature_importance table. Sorted descending. Responds to Top 5 / All Factors bookmark toggle.
- **Donut Chart:** Importance grouped by category (Facility, Behavioral, Economic, Temporal, Weather, Congestion). Shows that Facility category alone accounts for 75.18% of model importance.
- **KPI Cards:** Model R^2 (0.9791), RMSE 2,718 vehicles/day, Top Factor Importance 75.18%, Model: H2O Stacked Ensemble.

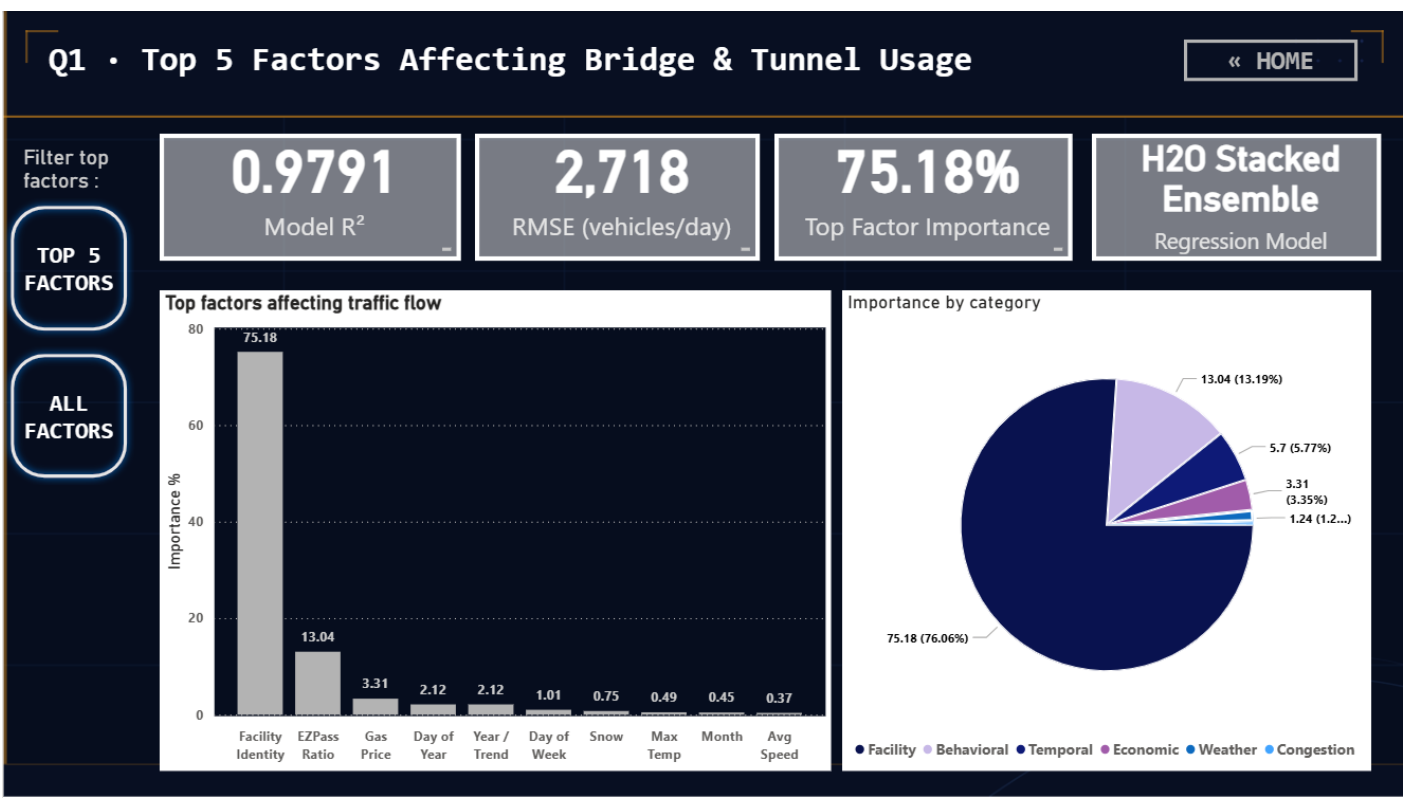


Figure 2.2 — Q1: Top 5 Factors Affecting Bridge & Tunnel Usage

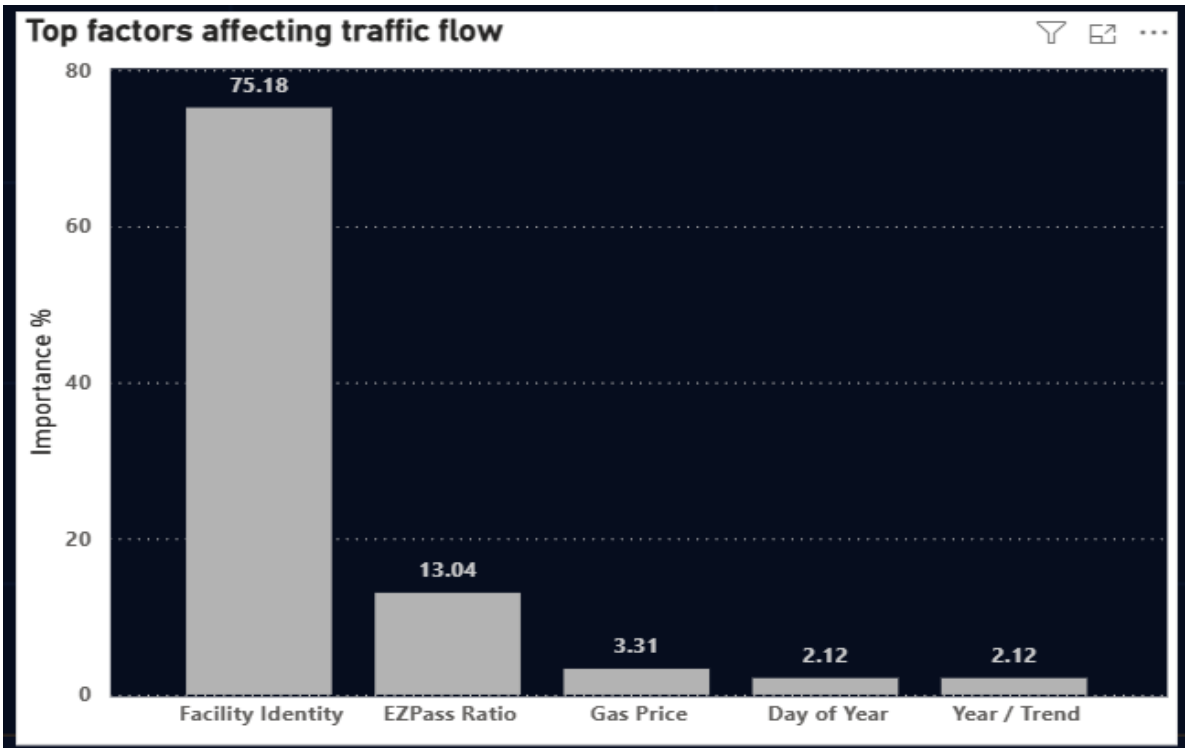


Figure 2.3 — Q1: Top 5 Factors view — Facility Identity dominates at 75.18%

2.4 Q2 — Toll Violations by Time Period & Facility

This page answers the second business question using results from the **H2O AutoML Classification Gradient Boosting model** (AUC = 0.9985, F1 = 0.9711, Recall = 96.79%). A facility slicer allows the company to filter all three charts to a specific crossing. Four KPI cards display classification model performance metrics.

- **Violations by Year (Bar Chart):** Total violations per year 2015–2024 from the main dataset. X-axis: Year, Y-axis: Total Violations measure. Shows peak violations in 2020–2021 and declining trend post-2022.
- **Violations by Facility (Bar Chart):** Total violations per crossing sorted descending. GWB Lower (17.7M) and GWB Upper (17.1M) have the highest cumulative violations. Bayonne (1.7M) the lowest.
- **Avg Violation Rate by Month (Line Chart):** Average violation rate by calendar month. Shows summer peak (June–August) with rates reaching 0.067, confirming seasonal enforcement challenges.
- **Facility Slicer:** Dropdown slicer on fac_b — filters all three charts simultaneously to a selected facility.

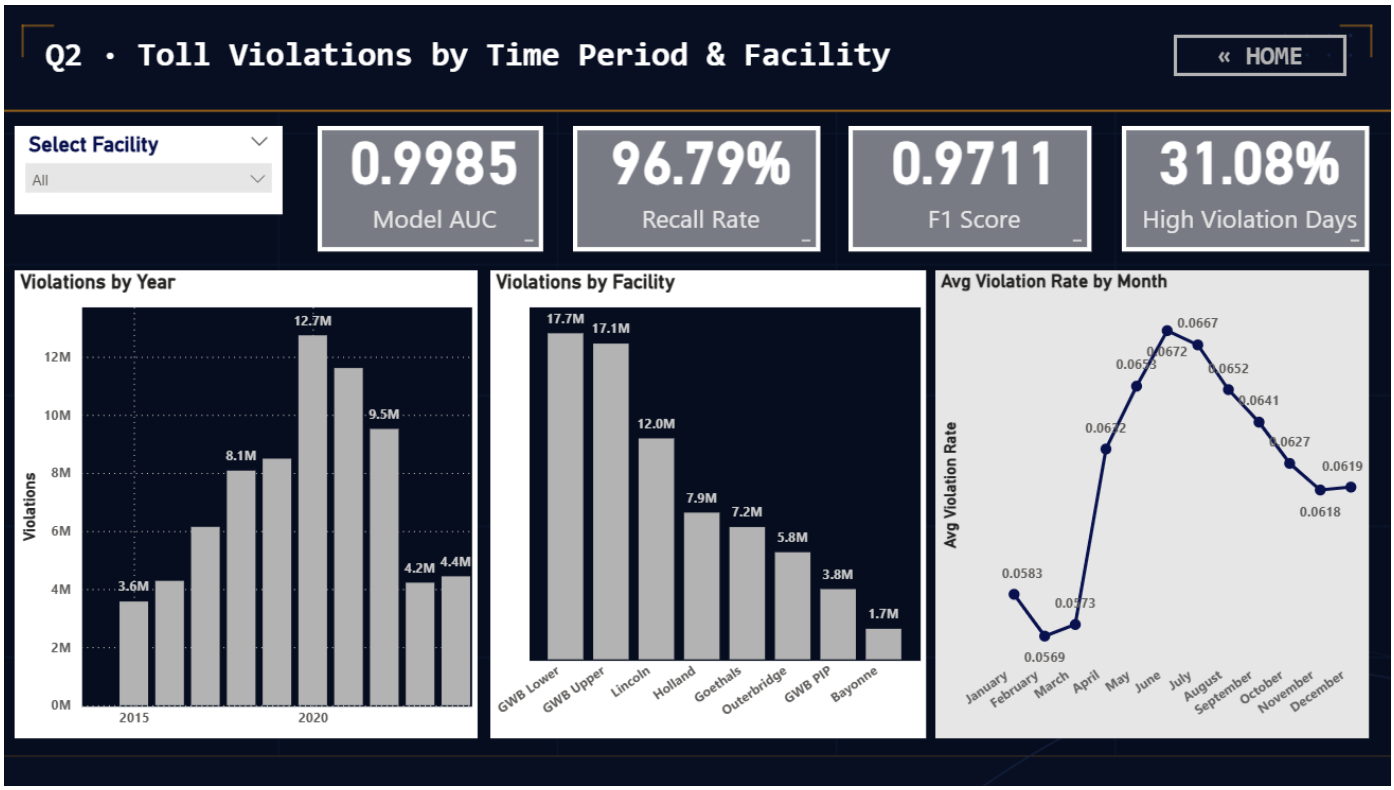


Figure 2.4 — Q2: Toll Violations by Time Period & Facility

2.5 Q3 — Busiest Times & Seasonal Traffic Patterns

This page answers the third business question using the main traffic dataset analyzed through the **H2O Regression Stacked Ensemble model** temporal features. Four charts together reveal seasonality, day-of-week patterns, COVID-19 impact, and facility-level trends. The Traffic by Facility Over Time chart uses the Combined_Traffic table which includes both actual data (2015–2024) and forecast data (2025–2032) for a complete picture.

- **Network Traffic by Year (Bar Chart):** Total vehicles across all 8 facilities per year. 2020 visually confirmed as the COVID-19 dip year (98M vs 128M in 2019). 2023 is the peak year at 129M vehicles.
- **Avg Traffic by Month (Line Chart):** Average daily vehicles per month showing the June–August summer peak (44.0K–44.9K) vs January trough (38.7K). Confirms Day of Year as a key regression feature.
- **Avg Traffic by Day of Week (Bar Chart):** Average vehicles per day of week. Friday and Saturday show the highest volumes (44K), Monday the lowest (42K), consistent with leisure travel patterns.
- **Traffic by Facility Over Time (Line Chart):** Per-facility yearly vehicle totals from 2015 to 2032 (actual + forecast). GWB Upper dominates but shows a declining trend. Bayonne shows the steepest growth trajectory.

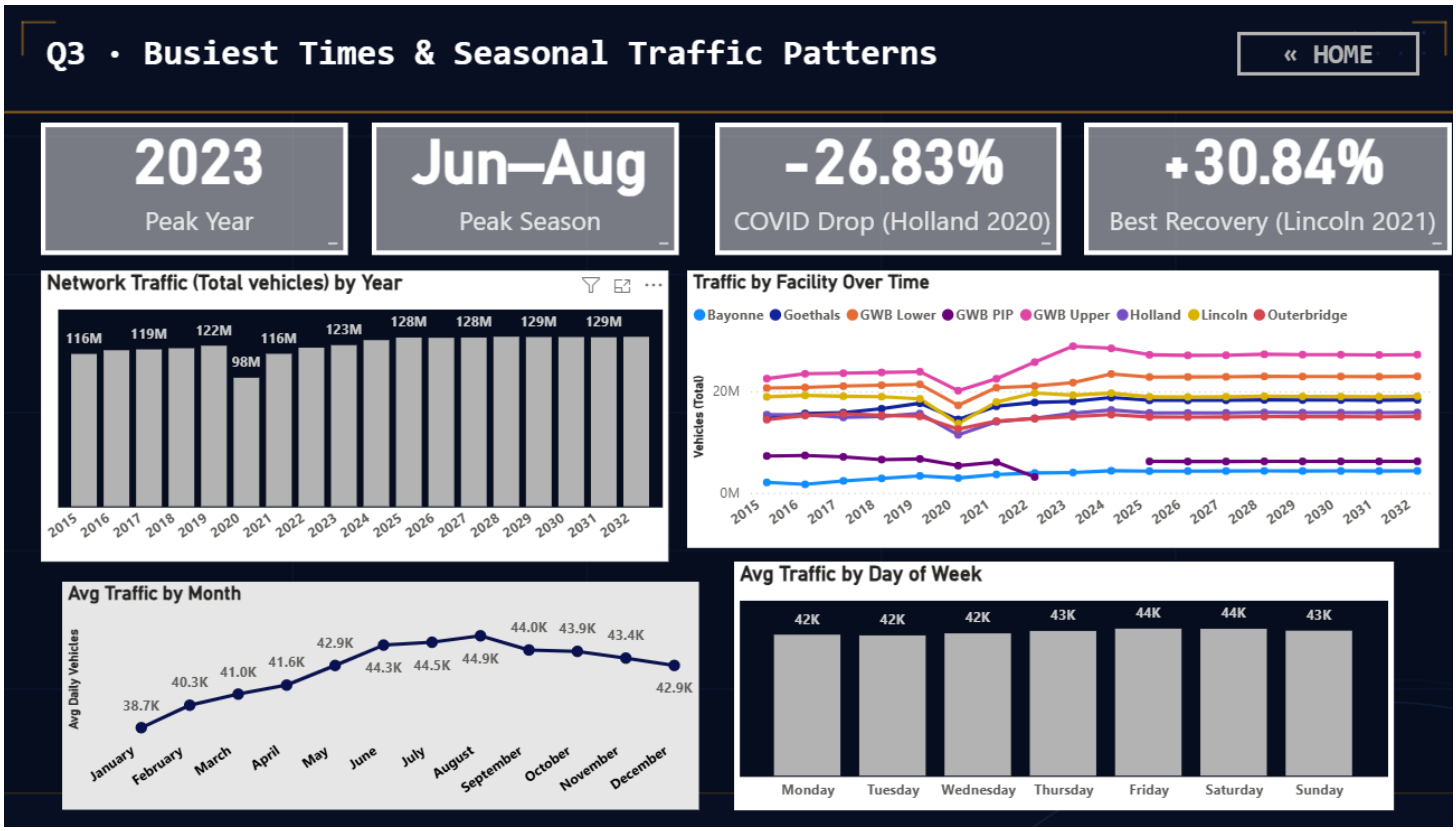


Figure 2.5 — Q3: Busiest Times & Seasonal Traffic Patterns

2.6 Q4 — Congestion Pricing Impact in 2025

This page answers the fourth business question by comparing traffic volumes across 2023, 2024, and 2025 per facility. The 2025 data comes from the AutoTS forecast table (full year projected values) appended into the Combined_Traffic table, replacing the incomplete January–May actual data. A data note is included below Chart 1 stating that GWB PIP data is unavailable for 2023–2024.

- **Traffic Volume: 2023 vs 2024 vs 2025 (Clustered Bar Chart):** Three grouped bars per facility showing year-over-year comparison. GWB Upper shows a clear declining trend from 2023 to 2025. GWB Lower shows an increasing trend — a potential congestion pricing beneficiary.
- **2025 Projected Traffic by Facility (Bar Chart):** Full year 2025 forecast from AutoTS model per facility. GWB Upper leads at 27.2M projected vehicles, Bayonne lowest at 4.3M. Gives management a full-year 2025 outlook without the limitation of partial actual data.
- **KPI Cards:** Congestion Pricing Start (Jan 2025), GWB Upper YoY Change 2024 (−1.46%), Highest Traffic Decline 2023–2025 (GWB Upper), Highest Volume Facility 2025 (GWB Lower).

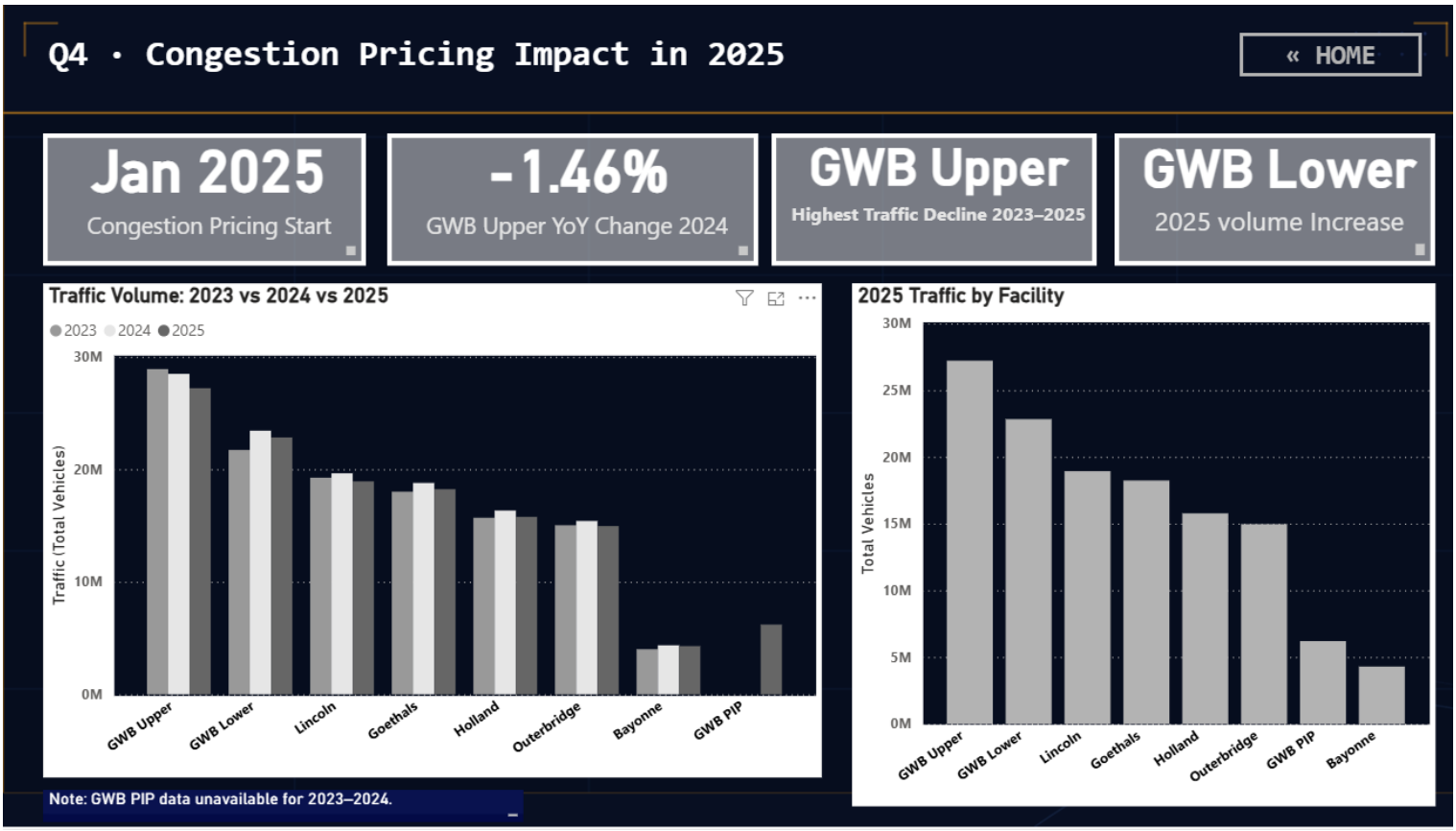
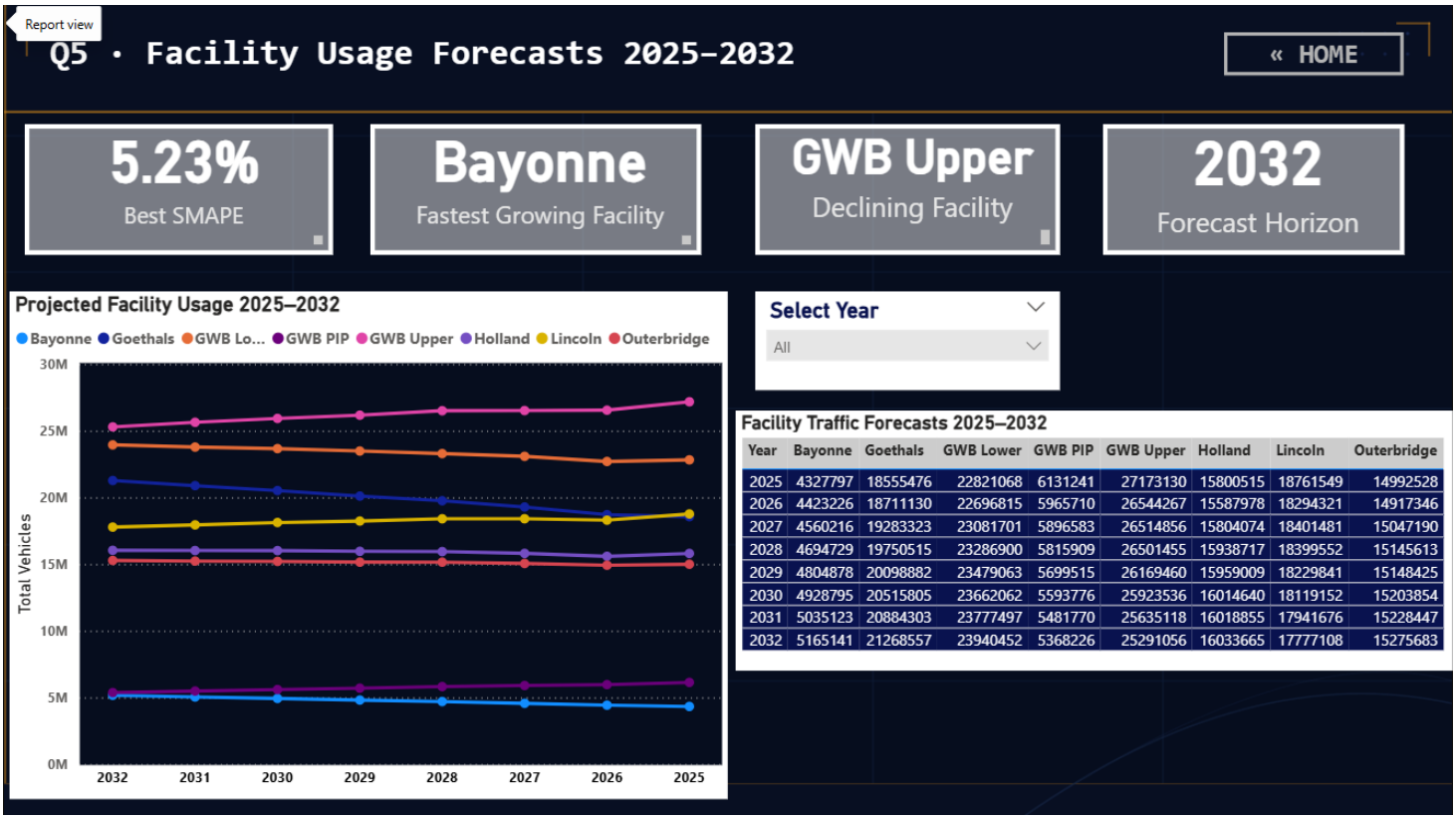


Figure 2.6 — Q4: Congestion Pricing Impact in 2025

2.7 Q5 — Facility Usage Forecasts 2025–2032

This page presents the AutoTS Horizontal Ensemble forecast results for all eight Port Authority facilities from 2025 through 2032. The forecast dataset was generated from the trained AutoTS model with forecast_length=2922 days. Two interactive slicers — Facility and Year — filter both the line chart and the forecast table simultaneously since both visuals draw from the same unpivoted forecast table.

- **Projected Facility Usage Line Chart:** Multi-series line chart showing yearly total vehicle forecasts per facility 2025–2032. GWB Upper (top line, ~27M declining) and Bayonne (bottom line, ~4.3M growing) show the most distinct trends. Y-axis range 0M–30M.
- **Facility Traffic Forecasts Table:** Matrix visual with Year as rows and facilities as columns. Values shown in millions (vehicles_millions calculated column). Allows management to read exact projected volumes for any facility-year combination.
- **Slicers:** Select Year (dropdown) —filter the line chart and table simultaneously. Selecting Bayonne isolates its growth trajectory; selecting GWB Upper reveals the projected 7% decline by 2032.
- **KPI Cards:** Best SMAPE 5.23% (AutoTS · Horizontal Ensemble), Fastest Growing Facility (Bayonne, 4.3M → 5.2M), Declining Facility (GWB Upper, 27.2M → 25.3M), Forecast Horizon (2032).



Year	Bayonne	Goethals	GWB Lower	GWB PIP	GWB Upper	Holland	Lincoln	Outerbridge
2025	4327797	18555476	22821068	6131241	27173130	15800515	18761549	14992528
2026	4423226	18711130	22696815	5965710	26544267	15587978	18294321	14917346
2027	4560216	19283323	23081701	5896583	26514856	15804074	18401481	15047190
2028	4694729	19750515	23286900	5815909	26501455	15938717	18399552	15145613
2029	4804878	20098882	23479063	5699515	26169460	15959009	18229841	15148425
2030	4928795	20515805	23662062	5593776	25923536	16014640	18119152	15203854
2031	5035123	20884303	23777497	5481770	25635118	16018855	17941676	15228447
2032	5165141	21268557	23940452	5368226	25291056	16033665	17777108	15275683

Figure 2.7 — Q5: Facility Usage Forecasts 2025–2032

Facility Traffic Forecasts 2025–2032

date	Bayonne	Goethals	GWB Lower	GWB PIP	GWB Upper	Holland	Lincoln	Outerbridge
2025	4327797	18555476	22821068	6131241	27173130	15800515	18761549	14992528
2026	4423226	18711130	22696815	5965710	26544267	15587978	18294321	14917346
2027	4560216	19283323	23081701	5896583	26514856	15804074	18401481	15047190
2028	4694729	19750515	23286900	5815909	26501455	15938717	18399552	15145613
2029	4804878	20098882	23479063	5699515	26169460	15959009	18229841	15148425
2030	4928795	20515805	23662062	5593776	25923536	16014640	18119152	15203854
2031	5035123	20884303	23777497	5481770	25635118	16018855	17941676	15228447
2032	5165141	21268557	23940452	5368226	25291056	16033665	17777108	15275683

Figure 2.8 — Q5: Facility Traffic Forecasts Table (in Millions)

2.8 Dashboard Design Decisions

The following design decisions were made to ensure the dashboard is professional, readable, and appropriate for management-level use:

- **Dark navy theme:** A consistent dark background (#060d1e) with amber (#c8821a) accent lines was used across all pages — matching the web application theme and providing strong contrast for chart readability in presentation settings.
- **KPI cards on every page:** Each question page leads with 4 KPI cards showing the most critical model metrics and findings. This allows management to absorb key numbers at a glance before examining the detailed charts.
- **Question labeled clearly on every page:** Each page title explicitly states the business question it answers, eliminating ambiguity about which visual corresponds to which requirement.

- **Facility slicer on Q2 and Q5:** Interactive slicers allow the Port Authority to filter data to specific crossings — enabling facility-level operational decisions without requiring separate reports.
- **Combined actual + forecast data on Q3:** The Traffic by Facility Over Time chart on Q3 shows both actual historical data (2015–2024) and AutoTS forecasts (2025–2032) in one continuous view, giving management a complete 17-year picture of each facility's trajectory.
- **No text annotations on charts:** Following professional dashboard standards, no explanatory text is placed inside chart areas. All context is provided through clear titles, axis labels, and KPI cards.

Section 3 — Web Application Deployment

This section documents the design, architecture, and deployment of an interactive web application built to communicate the findings of this project to the Port Authority of New York and New Jersey. The application serves as a live, browser-accessible analytics dashboard that directly answers each of the five business questions using pre-computed model outputs and the master dataset. It is publicly accessible at: **https://port-authority-ny-nj-traffic-flow-c.vercel.app**.

3.1 Application Overview

The web application is a single-page HTML dashboard that answers all five business questions posed by the Port Authority. It was built using plain HTML, CSS, and JavaScript — with no backend framework required — making it lightweight, portable, and fully deployable on Vercel without any server-side infrastructure. The application pulls all data in real time from a Supabase PostgreSQL database, which stores pre-computed analytics results derived from the master dataset and the three predictive models built during Phase 2.

The dashboard is organized around a left-side filter panel and a right-side question panel. Users select a business question from a dropdown menu, and the corresponding charts, KPI cards, and dynamic insights appear instantly. Filters for facility, year, and metric focus (Traffic Volume, Violations, or EZPass Ratio) allow users to drill into specific segments of the data without any technical knowledge.

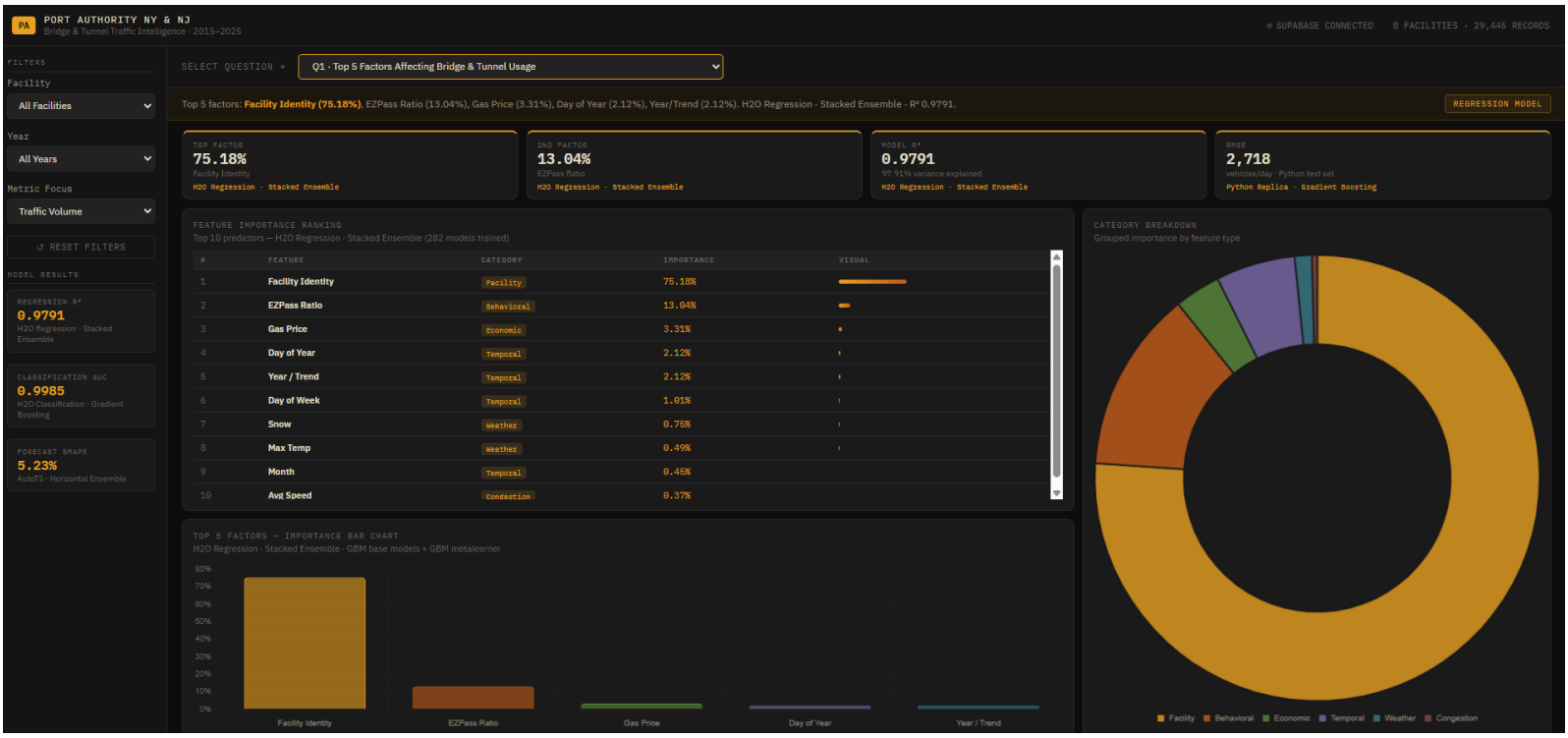


Figure 3.1 — Q1 Dashboard: Top 5 Factors Affecting Bridge & Tunnel Usage

3.2 Technology Stack

The application was built using the following technologies, selected for simplicity, cost-effectiveness, and ease of deployment:

Component	Technology	Purpose
Frontend	HTML / CSS / JavaScript	Single-page dashboard with no framework dependencies
Charting	Chart.js v4.4.1	Interactive bar, line, and doughnut charts for all 5 questions
Database	Supabase (PostgreSQL)	Cloud database storing all pre-computed analytics tables
Data Pipeline	Python (pandas)	Pre-computation of all aggregated metrics from the master CSV
Hosting	Vercel	Zero-configuration deployment with global CDN and HTTPS
Source Control	GitHub	Version control and deployment trigger for Vercel

Chart.js was selected for its lightweight footprint and native support for responsive canvas-based charts. Supabase was chosen over a static JSON file because it provides a REST API that the dashboard queries on every page load, meaning the data can be updated without redeploying the application. Vercel was selected because it supports direct GitHub integration and deploys a static HTML file in under 30 seconds.

3.3 Data Architecture

All model outputs and aggregated metrics were pre-computed from the master dataset (Traffic_by_Facility_FINAL-1.csv, 29,445 rows) using Python scripts and uploaded to Supabase as structured relational tables. The dashboard never queries raw data — it reads from pre-aggregated tables only, ensuring fast load times and clean separation between the analytical pipeline and the presentation layer.

The following tables were created in Supabase to power the five dashboard sections:

Table Name	Rows	Purpose
feature_importance	10	Regression model feature importance rankings and scores
violations_by_year	11	Total violations and avg violation rate per year
violations_by_month	12	Avg violation rate by month across all facilities
violations_by_facility	8	Total violations and high-violation days per facility

traffic_by_year	11	Total network vehicles per year 2015–2025
traffic_by_month	12	Average daily vehicles per month across all facilities
traffic_by_dow	7	Average vehicles by day of week
traffic_by_facility_year	85	Yearly vehicle totals per facility for trend charts
traffic_by_fac_year_month	966	Monthly aggregates per facility per year for drill-down filters
traffic_by_fac_year_dow	595	Day-of-week aggregates per facility per year for filters
congestion_pricing	21	Traffic volumes for 2023, 2024, and 2025 per facility
forecasts	8	AutoTS model forecasts 2025–2032 per facility
ezpass_by_facility	8	Avg EZPass ratio per facility
ezpass_by_year	11	Avg EZPass ratio per year
ezpass_by_month	12	Avg EZPass ratio per month

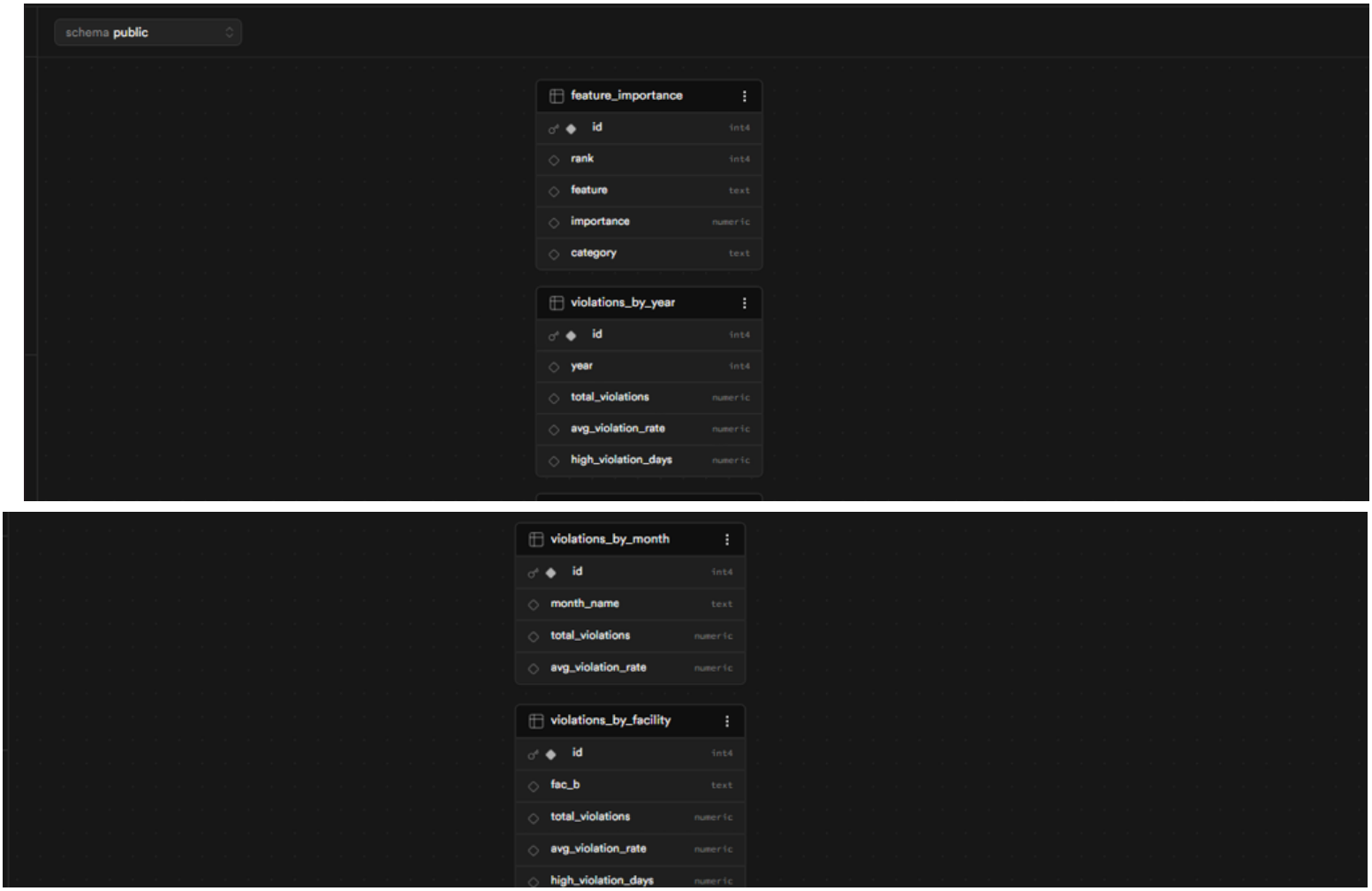


Figure 3.2 — Supabase Database Schema showing all analytics tables

3.4 Dashboard Features & Business Question Mapping

Each section of the dashboard is explicitly mapped to one of the five business questions. The question dropdown in the top bar makes this mapping visible to all users. Below is a description of what each section displays and how it answers the corresponding question.

3.4.1 Q1 — Top 5 Factors Affecting Bridge & Tunnel Usage

This section displays the results of the H2O Regression Stacked Ensemble model. The main panel contains a feature importance ranking table showing all 10 predictors with their importance scores, categories, and visual bars. Below the table, a horizontal bar chart highlights the top 5 factors. A doughnut chart on the right shows how importance is distributed across feature categories (Facility, Behavioral, Economic, Temporal, Weather, Congestion). KPI cards display the top factor (75.18% — Facility Identity), the second factor (13.04% — EZPass Ratio), the model R^2 (0.9791), and the RMSE (2,718 vehicles/day).

FEATURE IMPORTANCE RANKING
Top 10 predictors — H2O Regression · Stacked Ensemble (282 models trained)

#	FEATURE	CATEGORY	IMPORTANCE	VISUAL
1	Facility Identity	Facility	75.18%	
2	EZPass Ratio	Behavioral	13.04%	
3	Gas Price	Economic	3.31%	
4	Day of Year	Temporal	2.12%	
5	Year / Trend	Temporal	2.12%	
6	Day of Week	Temporal	1.01%	
7	Snow	Weather	0.75%	
8	Max Temp	Weather	0.49%	
9	Month	Temporal	0.45%	
10	Avg Speed	Congestion	0.37%	

Figure 3.3 — Feature Importance Ranking from H2O Regression · Stacked Ensemble

Figure 3.4 — Top 5 Factors Importance Bar Chart

3.4.2 Q2 — Toll Violations by Time Period & Facility

This section displays results from the H2O Classification Gradient Boosting model. Three charts are shown simultaneously: total violations by year, total violations by facility, and average violation rate by month. KPI cards show the model AUC (0.9985), recall (96.79%), the proportion of high-violation days (31.08%), and the F1 score (0.9711). When the Metric Focus filter is switched to EZPass Ratio,

all three charts update to show EZPass adoption trends instead, allowing users to see the inverse relationship between EZPass adoption and violation rates.

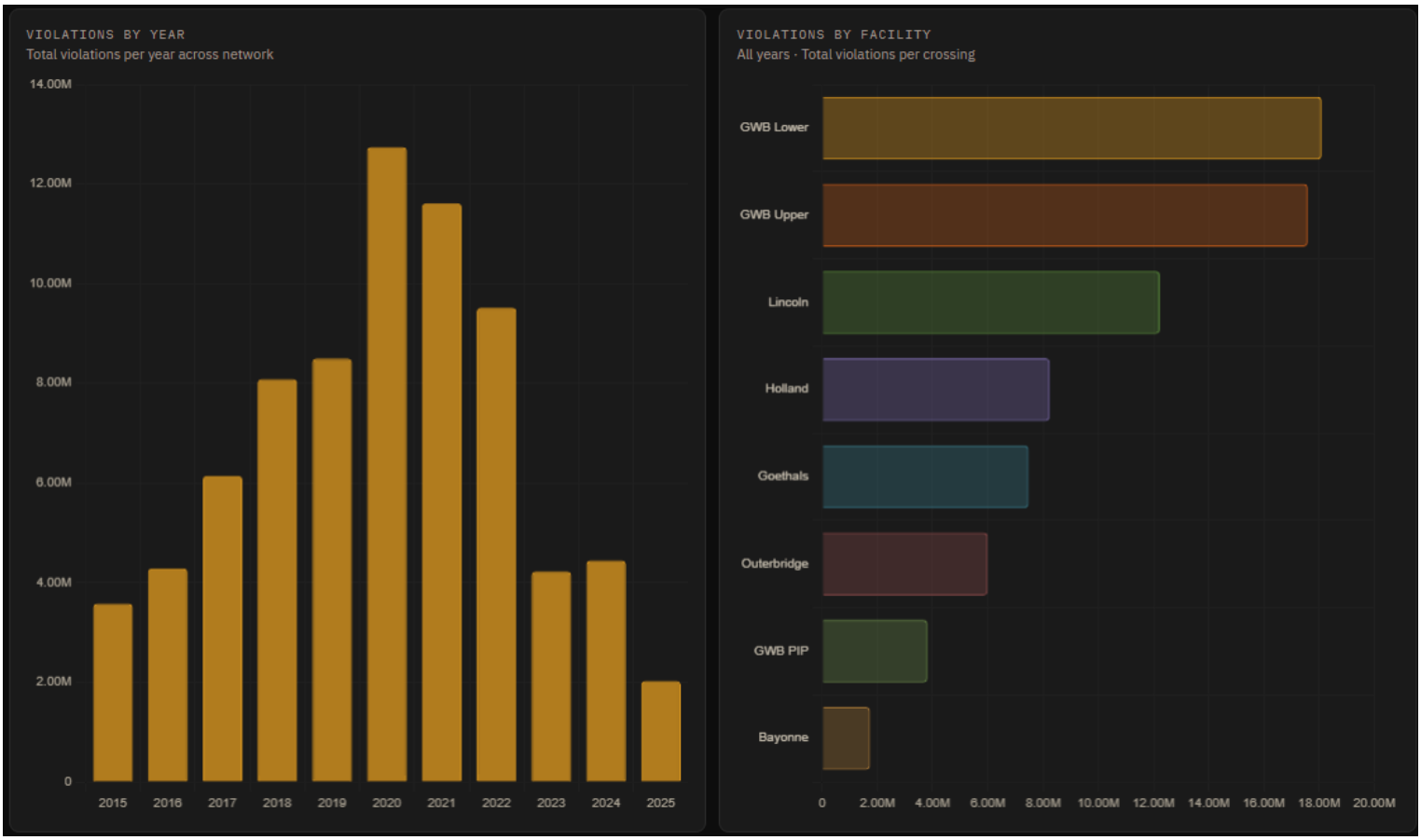


Figure 3.5 — Q2 Dashboard: Toll Violations by Time Period and Facility

3.4.3 Q3 — Busiest Times & Seasonal Traffic Patterns

This section addresses seasonality, day-of-week patterns, and year-over-year traffic trends. Four charts are displayed: network traffic by year (with 2020 highlighted in red to mark the COVID-19 impact), average traffic by month (revealing the June–August summer peak), traffic by day of week, and a multi-series line chart showing each facility's traffic trend from 2015 to 2025. All four charts respond to the facility and year filters — selecting a specific facility isolates its data across all charts simultaneously.

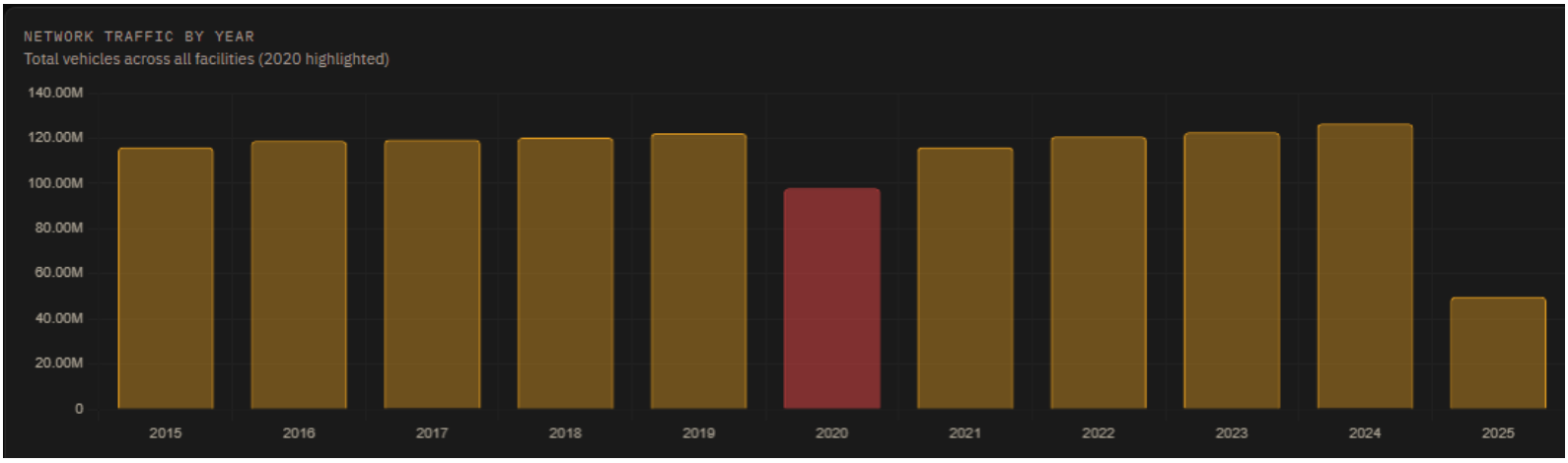


Figure 3.6 — Network Traffic by Year showing COVID-19 dip in 2020

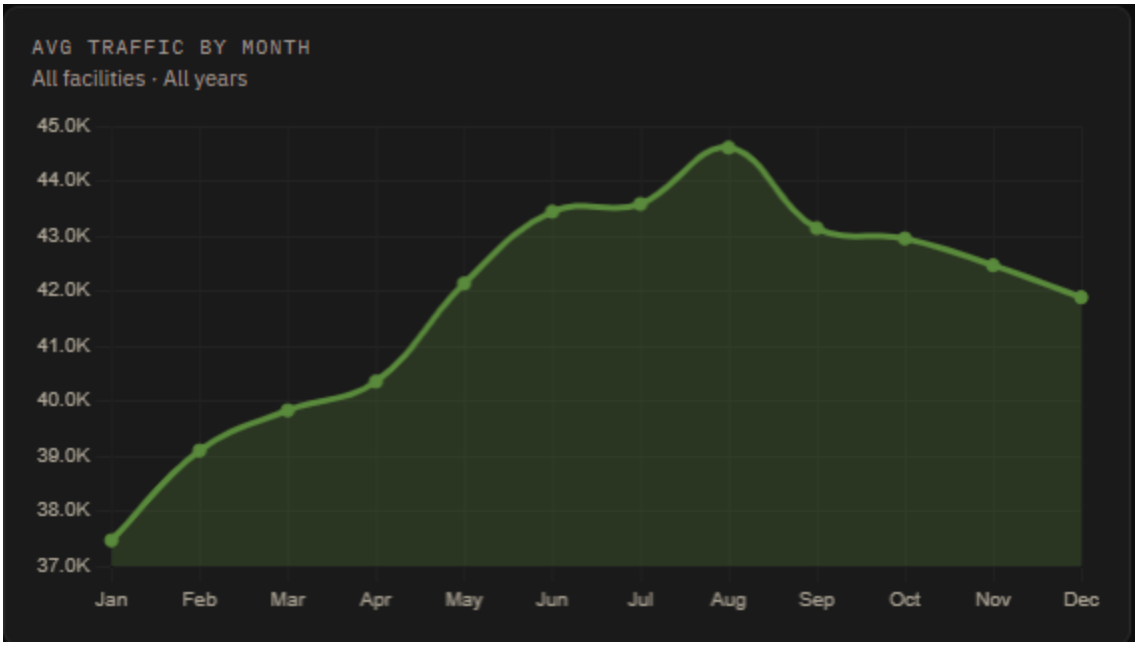


Figure 3.7 — Average Daily Traffic by Month showing June–August peak

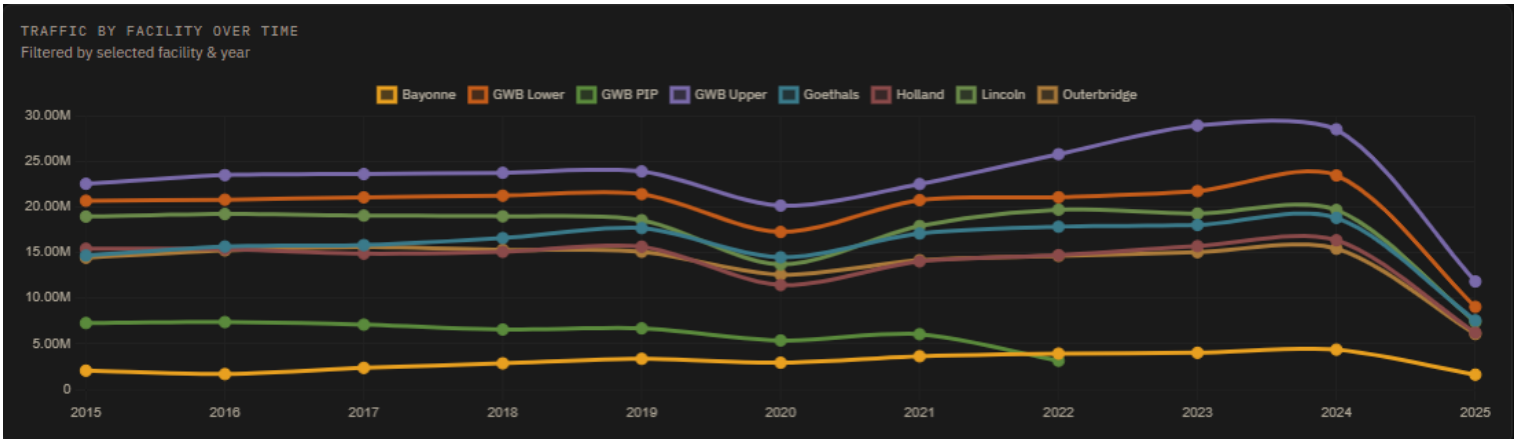


Figure 3.8 — Traffic by Facility Over Time 2015–2025

3.4.4 Q4 — Congestion Pricing Impact in 2025

This section compares annualized 2025 traffic volumes against 2023 and 2024 baselines per facility. The 2025 data covers January through May only and is annualized by multiplying by 365/150. A grouped bar chart shows the three-year comparison across all eight facilities. A second chart shows the percentage change from 2024 to 2025 per facility, with green bars indicating traffic gains and red bars indicating losses. GWB Upper — the only facility with a negative growth rate in 2024 (−1.46%) — appears prominently in the analysis.

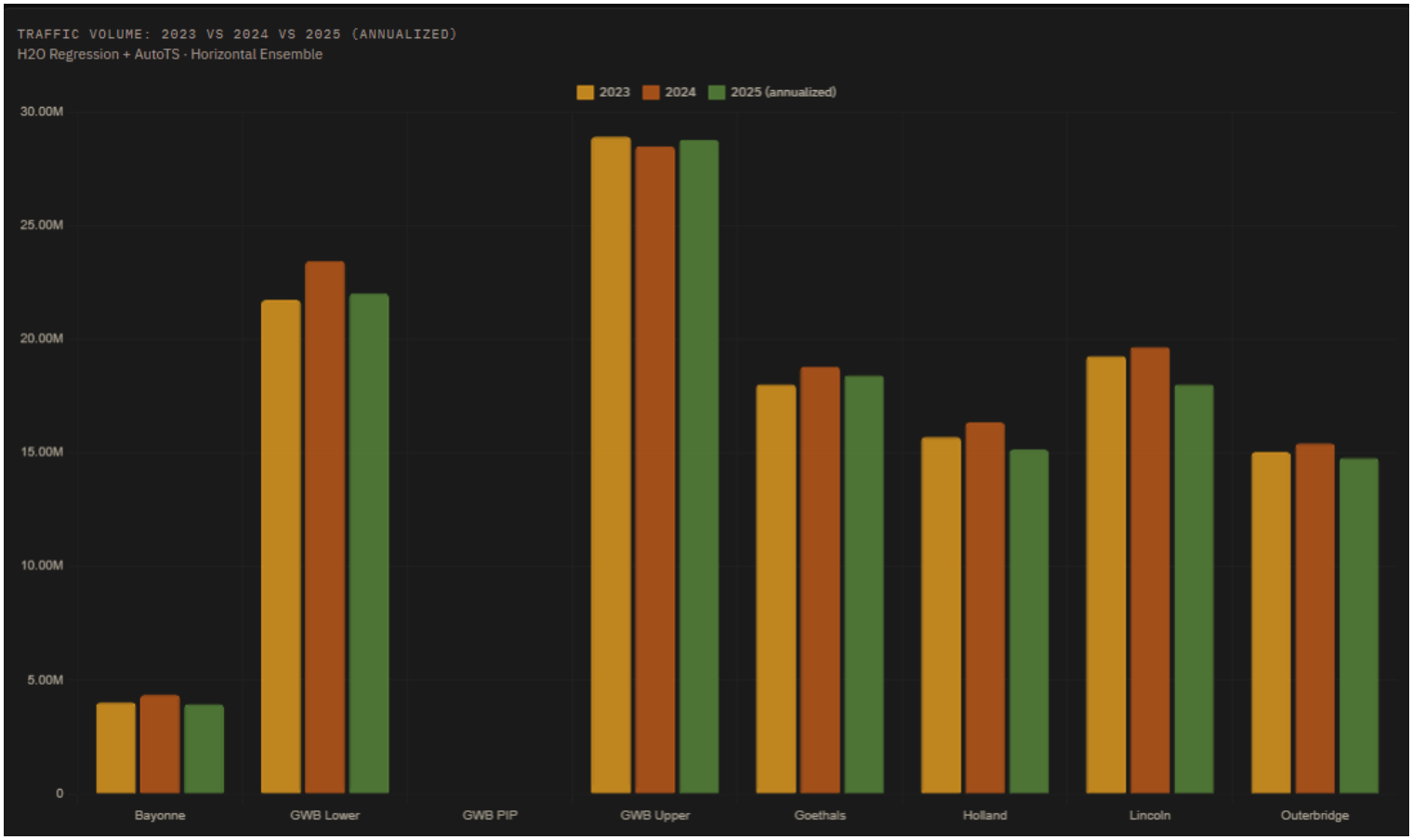


Figure 3.9 — Traffic Volume Comparison across 2023, 2024, and 2025 by Facility

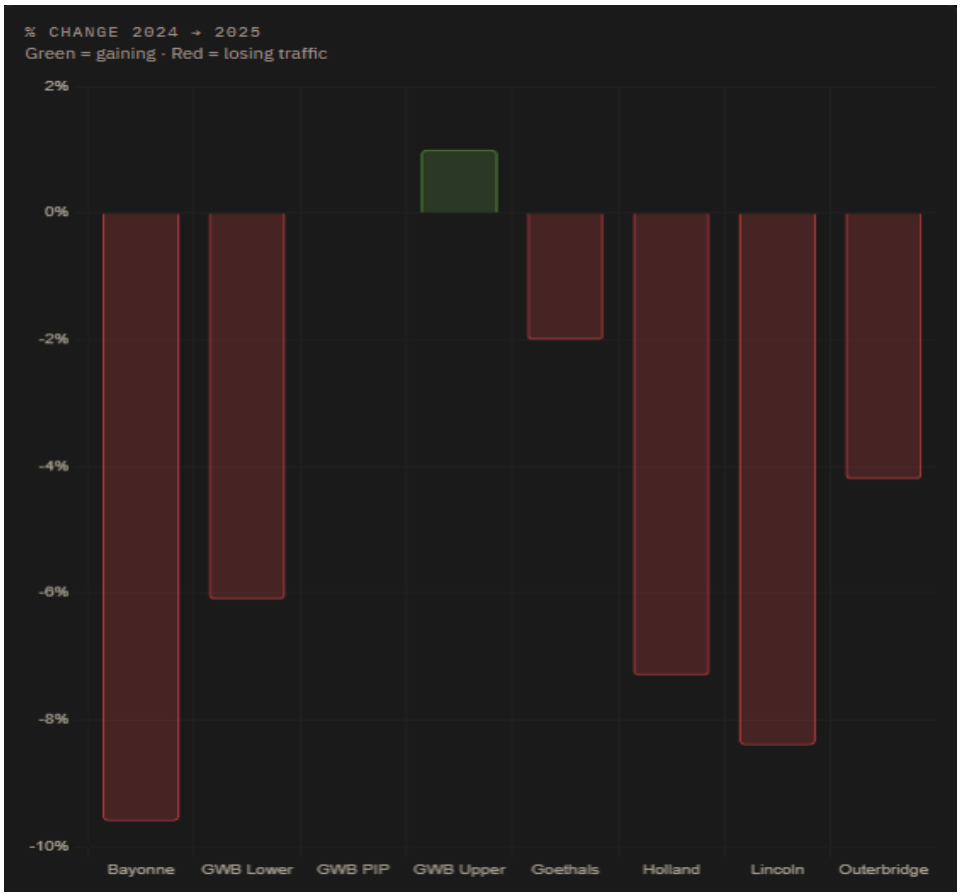


Figure 3.10 — Percentage Change 2024 to 2025 per Facility

3.4.5 Q5 — Facility Usage Forecasts to 2032

This section presents the AutoTS Horizontal Ensemble forecast results for all eight Port Authority facilities from 2025 through 2032. A multi-series line chart displays the projected yearly vehicle counts per facility across the 8-year horizon. A data table alongside the chart shows the exact forecast values in millions of vehicles. The facility filter on the left panel isolates individual facility forecasts. KPI cards identify the model (AutoTS Horizontal Ensemble), the best SMAPE (5.23%), the fastest-growing facility (Bayonne, +19.3%), and the declining facility (GWB Upper, −7.0%).

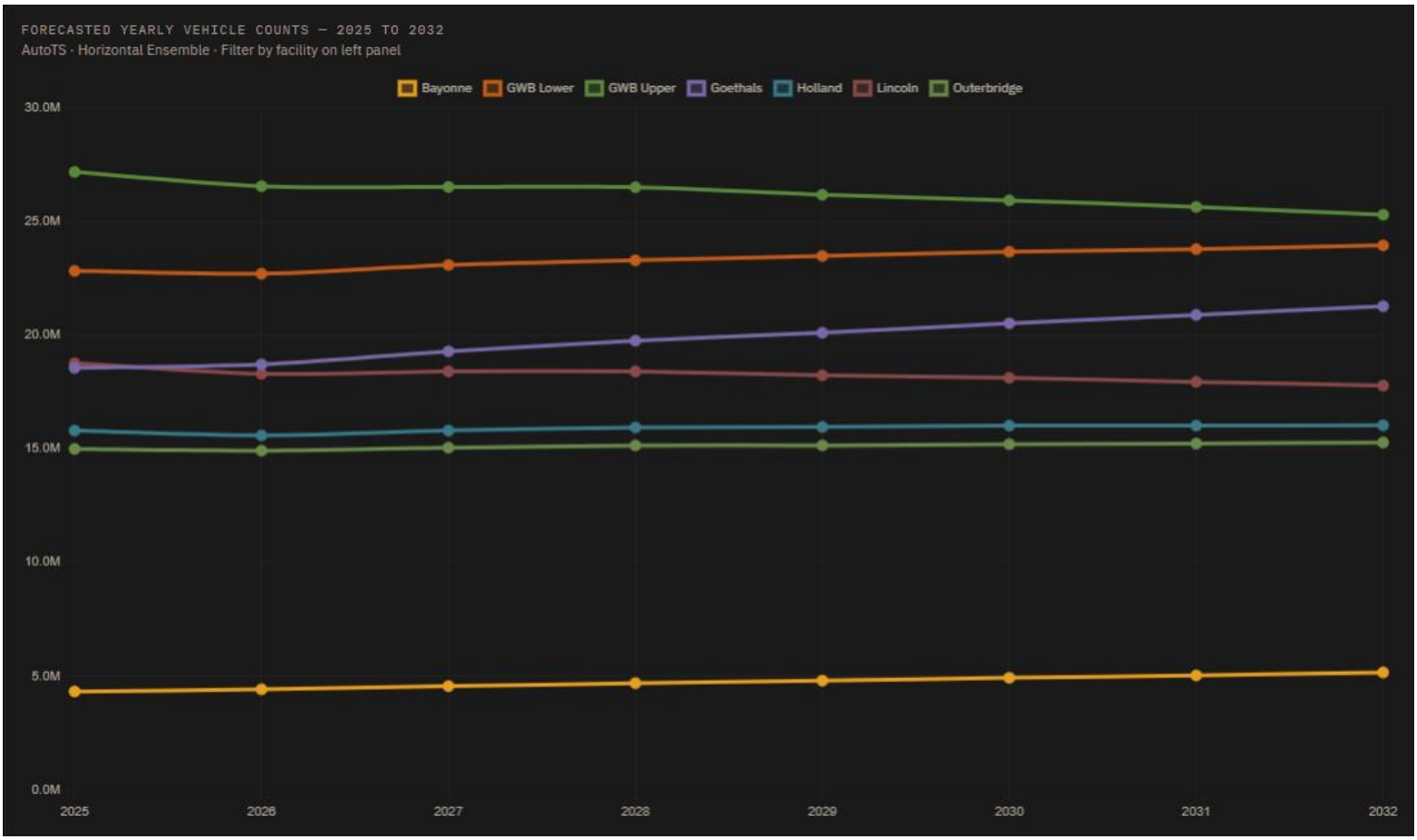


Figure 3.11 — AutoTS Horizontal Ensemble Forecasts 2025–2032

3.5 Interactive Filter System

A key feature of the dashboard is its filter system, which allows Port Authority management to explore the data interactively without technical assistance. Three filters are available on the left panel:

- **Facility:** Filters all charts to a specific crossing — Bayonne, GWB Lower, GWB PIP, GWB Upper, Goethals, Holland, Lincoln, or Outerbridge. When a facility is selected, all charts in Q2, Q3, and Q5 update to show only that facility's data.
- **Year:** Filters the data to a specific year. Selected years are highlighted in charts while other years are dimmed, making year-specific patterns immediately visible without losing context.
- **Metric Focus:** Switches the charts between three analytical lenses — Traffic Volume (default), Violations, and EZPass Ratio. This allows users to view the same time dimensions through different operational perspectives without navigating away from the current question.

A dynamic insight strip below the question dropdown updates automatically based on the active filters, providing a plain-language interpretation of what the selected data shows. For example, selecting GWB Upper in Q4 triggers the insight: "GWB Upper is the only facility with negative 2024 growth (−1.46%), consistent with early congestion pricing anticipation before January 2025 launch."

3.6 Deployment Pipeline

The application was deployed using the following pipeline:

- **Step 1 — Data Pre-computation:** Python scripts read the master CSV and computed all aggregated metrics, exporting them as SQL INSERT statements.
- **Step 2 — Database Setup:** Sixteen tables were created in Supabase using SQL DDL statements executed via the Supabase SQL Editor. All pre-computed data was inserted into the respective tables.
- **Step 3 — Application Build:** The dashboard was built as a single index.html file containing all HTML structure, CSS styling, and JavaScript logic. The Supabase URL and anon public key are embedded directly in the JavaScript, enabling direct REST API calls from the browser.
- **Step 4 — GitHub Repository:** The index.html file was uploaded to a GitHub repository named PortAuthority-NY-NJ-TrafficFlow_CapstoneProject under the account Meghana-l.
- **Step 5 — Vercel Deployment:** The GitHub repository was imported into Vercel. Vercel automatically detected the static HTML file, built the project with no configuration, and deployed it to a global CDN. The production URL was generated within 30 seconds of deployment.

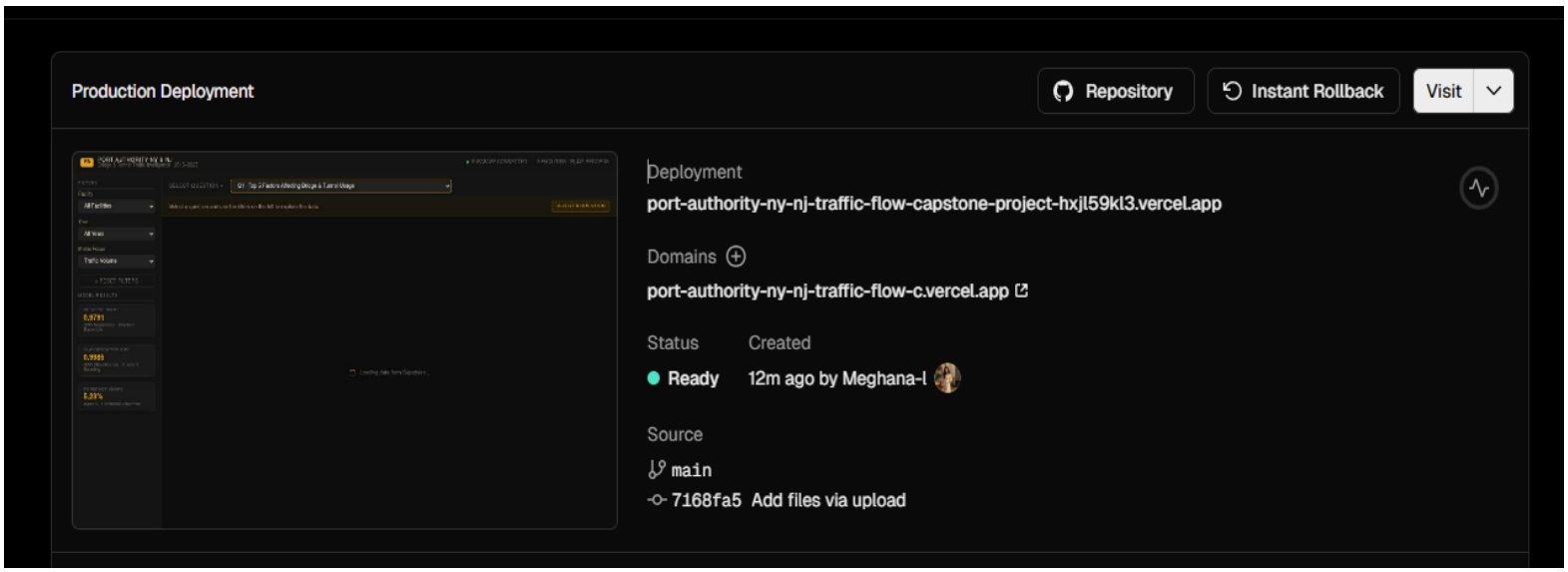


Figure 3.12 — Vercel Production Deployment confirming live status

The live application is accessible at: **https://port-authority-ny-nj-traffic-flow-c.vercel.app**

Any update to the GitHub repository's main branch automatically triggers a new Vercel deployment, ensuring the dashboard always reflects the latest version of the code.

3.7 Design Decisions

Several deliberate design decisions were made to ensure the application is appropriate for management-level use at the Port Authority:

- **No scrolling:** The entire dashboard fits within a standard laptop screen. All charts, KPI cards, and filters are visible without vertical scrolling, enabling efficient review during presentations or meetings.
- **Model attribution on every card:** Each KPI card and chart explicitly identifies which model produced the displayed result — H2O Regression · Stacked Ensemble, H2O Classification · Gradient Boosting, or AutoTS · Horizontal Ensemble. This prevents confusion about which analytical method supports each finding.
- **Model type tag:** A tag in the top-right corner of the insight strip identifies the model category for the active question — REGRESSION MODEL, CLASSIFICATION MODEL, TIME SERIES MODEL, or REGRESSION + TIME SERIES — making the analytical context immediately clear.

- **Dark theme:** A dark amber-on-black color scheme was chosen for visual clarity during presentations in darkened rooms, consistent with operational control room aesthetics common in transportation management contexts.
- **Pre-computed data only:** The dashboard does not run models live. All results are pre-computed and stored in Supabase. This ensures sub-second load times and eliminates dependency on compute resources for serving the application.

3.8 Future Scope

While the current application pulls from pre-computed analytics stored in a Supabase cloud database, the architecture is designed to support live data integration with minimal modification. Should the Port Authority provide access to a real-time traffic API — or begin publishing daily facility-level vehicle counts and violation data through an open data portal — the dashboard can be updated to query that API directly, replacing the static pre-computed tables with live data feeds. In this scenario, all five business question dashboards, the forecast comparisons, and the model-driven insights would update automatically as new data flows in, transforming the current reporting tool into a fully operational, real-time traffic intelligence platform. The Supabase database, Vercel deployment infrastructure, and modular JavaScript architecture of the application are all built to support this extension without a complete rebuild.

Section 4 — Recommendations to The Port Authority

The following recommendations are addressed directly to the management of the Port Authority of New York and New Jersey. They are derived from the findings of the three predictive models developed in this project — the H2O AutoML Regression Stacked Ensemble, the H2O AutoML Classification Gradient Boosting model, and the AutoTS Horizontal Ensemble time series model — applied to 29,445 facility-day records across eight crossings from 2015 through 2025. These recommendations are organized around five areas: critical factors requiring immediate attention, operational improvements, additional data to collect, exogenous factors to incorporate, and broader strategic recommendations.

4.1 Critical Factors Requiring Immediate Attention

Facility Identity and Structural Differentiation. The regression model identified facility identity as the single most powerful predictor of daily traffic volume, accounting for 75.18% of model importance. This finding has a direct operational implication: the Port Authority's eight crossings are not interchangeable — they serve structurally different driver populations, route purposes, and volume regimes. GWB Upper processes 28M+ vehicles per year while Bayonne processes 4.3M. Investment, staffing, and capacity decisions must be made at the facility level, not the network level. A one-size-fits-all operational policy will systematically underserve high-volume crossings and over-resource low-volume ones.

EZPass Adoption as a Revenue and Compliance Lever. EZPass Ratio is the second most important predictor of traffic volume (13.04% importance) and directly correlates with lower violation rates. GWB PIP (86.9% EZPass adoption) and Outerbridge (86.4%) have the lowest violation rates in the network. GWB Lower (77.9%) and GWB Upper (76.9%) have the lowest adoption and the highest violation rates. The Port Authority should treat EZPass penetration as a key performance indicator — setting facility-level adoption targets and deploying incentive programs (discounted rates, loyalty schemes) specifically at low-adoption facilities. Every percentage point of EZPass adoption gained translates directly into reduced enforcement costs and improved revenue predictability.

Congestion Pricing Diversion — Monitor Closely in 2025. GWB Upper is the only facility that showed negative year-over-year growth in 2024 (−1.46%) — before NYC's congestion pricing policy even launched on January 1, 2025. This anticipatory behavioral shift, combined with the AutoTS model's projection of a 7% decline at GWB Upper by 2032, indicates that congestion pricing is already reshaping network traffic patterns. The Port Authority must establish a real-time monitoring protocol for 2025 full-year data to determine whether the diversion is structural (requiring capacity and resource reallocation) or temporary (requiring only short-term operational adjustment).

4.2 Operational Improvements

Deploy the Classification Model for Proactive Enforcement Scheduling. The H2O Classification Gradient Boosting model achieves an AUC of 0.9985 and a recall rate of 96.79% — meaning it correctly identifies nearly all high-violation days before they occur, with only a 1.02% false alarm rate. The Port Authority should integrate this model into its enforcement scheduling workflow. By predicting which facility-days will exceed the 6.18% violation rate threshold, enforcement resources can be pre-positioned at the right crossing on the right day rather than deployed reactively. This shift from reactive to predictive enforcement has the potential to significantly reduce the 31.08% of facility-days currently classified as high-violation.

Reallocate Resources from GWB Upper to Bayonne and Goethals. The AutoTS forecasts project GWB Upper to decline 7% by 2032 while Bayonne grows 19.3% and Goethals grows 14.6%. These diverging trajectories suggest that the current resource allocation — which reflects historical volume distribution — will become increasingly misaligned with future demand. The Port Authority should begin planning capacity and staffing adjustments that reduce GWB Upper overhead and expand Bayonne and Goethals infrastructure in anticipation of sustained growth at those crossings.

Address Seasonal Enforcement Gaps. Violation rates peak in summer months (June: 6.53%, July: 6.72%, August: 6.67%) — precisely the period of highest traffic volume. This combination of high volume and high violation rate creates the worst operational conditions of the year. The Port Authority should plan for augmented enforcement staffing and technology deployment (additional camera coverage, mobile enforcement units) specifically during June through August at all eight facilities.

4.3 Additional Data to Collect in the Future

The current dataset — while rich — has several gaps that limit the analytical depth achievable with the current models. The Port Authority should prioritize collecting the following data going forward:

- **Real-time traffic speed and congestion data:** The avg_speed column in the current dataset has significant null values (approximately 40% missing), limiting its use as a congestion predictor. Comprehensive, real-time speed data from all 8 facilities would substantially improve the regression model's ability to predict volume under congestion conditions.
- **Complete hourly or sub-daily traffic records:** The current dataset operates at a daily facility-level granularity. Hourly traffic counts would enable time-of-day analysis — identifying morning peak, evening peak, and overnight patterns that are invisible in daily aggregates. This is particularly valuable for tunnel facilities where peak-hour congestion is a daily operational challenge.
- **Toll revenue by vehicle class and payment method:** The current dataset tracks vehicle counts and violation rates but does not include revenue figures. Adding toll revenue by vehicle class (Class 1 through Class 6) and payment method (cash, EZPass, violation) would enable revenue forecasting models and

allow the Port Authority to quantify the financial impact of EZPass expansion campaigns.

- **Incident and maintenance event logs:** Traffic disruptions caused by accidents, lane closures, and scheduled maintenance are not captured in the current dataset. These events create anomalous volume spikes or drops that the models interpret as noise rather than assignable causes. Linking incident logs to traffic records would improve model accuracy and enable disruption impact quantification.
- **Competitor route data:** Traffic volumes on competing routes — particularly the Tappan Zee/Mario Cuomo Bridge, the Verrazano-Narrows Bridge, and key NJ Turnpike interchanges — would allow the Port Authority to measure cross-facility diversion effects directly rather than inferring them from volume changes at its own crossings. This data is particularly critical for tracking congestion pricing diversion in 2025 and beyond.

4.4 Exogenous Factors to Incorporate in Future Analyses

Several external factors that measurably affect bridge and tunnel usage are not currently captured in the dataset. Incorporating these into future model iterations would meaningfully improve predictive accuracy:

- **Remote work prevalence index:** The COVID-19 pandemic demonstrated that remote work adoption is the single most powerful external driver of bridge and tunnel volume — the 2020 network-wide decline of 20.5% was almost entirely attributable to remote work and mobility restrictions. A monthly remote work index (available from sources such as LinkedIn or the Bureau of Labor Statistics) would allow the model to quantify and forecast the ongoing effect of hybrid work policies on commuter traffic, particularly at tunnel facilities serving Manhattan workers.
- **NYC congestion pricing zone toll rates:** The January 2025 congestion pricing launch introduced a new toll variable that was not present in the training data. Future model iterations should include the congestion pricing toll rate as a feature — allowing the regression model to directly quantify the per-dollar impact of Manhattan entry costs on Port Authority crossing volumes. Any future congestion pricing rate adjustments should trigger immediate model retraining.
- **Transit service disruptions and MTA capacity:** When subway and commuter rail service is disrupted, a measurable number of transit riders switch to vehicle crossings. Incorporating MTA service reliability metrics and scheduled major line disruptions as features would improve the model's ability to predict unusual volume spikes at tunnel facilities.
- **Regional economic indicators:** Employment levels in Manhattan, NJ GDP growth, and consumer confidence indices are structural drivers of long-term commuter volume. These macro variables would improve the AutoTS model's long-range forecast accuracy beyond 2027 — a horizon at which the current model's SMAPE begins to widen.
- **Weather severity beyond snowfall and temperature:** The current dataset includes snow depth, snowfall, max/min temperature, wind speed, and precipitation. However, extreme weather events — hurricanes, ice storms, and heat emergencies — cause non-linear traffic responses that are not captured by these continuous variables. Adding a categorical severe weather flag that distinguishes routine bad weather from emergency-level events would improve the model's handling of weather-driven volume anomalies.

4.5 Additional Strategic Recommendations

Retrain Models Annually with Updated Data. The three models developed in this project were trained on data through December 2024. As 2025 and subsequent years produce new observations — particularly under the novel conditions of NYC congestion pricing — the models will need to be retrained to maintain their predictive accuracy. The Port Authority should establish an annual model refresh cycle: append each new calendar year's data, retrain the AutoML and AutoTS models, evaluate performance against the previous year's predictions, and update the Power BI dashboard and web application with the new model outputs. The infrastructure developed in this project — the Supabase database, the Vercel-deployed dashboard, and the Power BI file — is designed to support this update process with minimal friction.

Use the Forecasts as a Capacity Planning Input. The AutoTS model's facility-level forecasts through 2032 should be used directly as inputs to the Port Authority's capital planning process. Bayonne's projected 19.3% growth and Goethals' 14.6% growth suggest that both crossings will experience meaningful capacity pressure by the late 2020s. Infrastructure projects — lane expansions, toll plaza upgrades, approach road improvements — require 3–7 year planning horizons. Beginning the assessment process now, using the 2025–2032 forecasts as the demand baseline, would allow the Port Authority to get ahead of capacity constraints rather than responding to them reactively.

Expand the Dashboard for Operational Use. The interactive web application and Power BI dashboard developed in this project were designed as analytical reporting tools. With additional development — particularly the integration of real-time traffic feeds from the 511NY API or Port Authority operational systems — the dashboard could be extended into a live operational monitoring platform. Real-time violation rate tracking, daily volume vs. forecast comparison, and automated alerts for facilities approaching high-violation thresholds are all achievable extensions of the current architecture. The Supabase database is already structured to support real-time data ingestion, and the Vercel deployment infrastructure can handle the additional load.

Share Model Findings with Peer Agencies. The Port Authority operates in a regional ecosystem alongside the MTA, NJ Transit, the Triborough Bridge and Tunnel Authority, and the New York State Thruway Authority. The congestion pricing diversion findings — particularly the evidence of traffic shifting from GWB Upper to GWB Lower and Bayonne — have implications for all of these agencies' planning processes. Sharing the model results and forecast data through existing inter-agency coordination mechanisms would enable system-level transportation planning that accounts for cross-network substitution effects, rather than each agency optimizing in isolation.